OFFLINE EDITION

Spark Notebook

8 topics · 17 lessons · quiz answers at the end of each lesson

ARTIFICIAL INTELLIGENCE

- How machines learn
- How chatbots work

ENGLISH

- Tempos verbais sem confusão
- Falsos amigos
- Preposições e verbos do dia a dia

ELECTRICITY

- Voltage, current & resistance
- Circuits at home

ELECTRONICS

- Components you meet first
- Signals & microcontrollers

ENGINEERING

- Forces, levers & gears
- How structures stand up

PHYSICS

- Energy & heat

- Waves, sound & light

HISTORY

- Turning points of humanity
- Portugal and the Age of Discovery

WORLD

- How the planet works
- People and places

AI

Artificial Intelligence

How machines learn and what's really inside a chatbot, explained without maths homework.

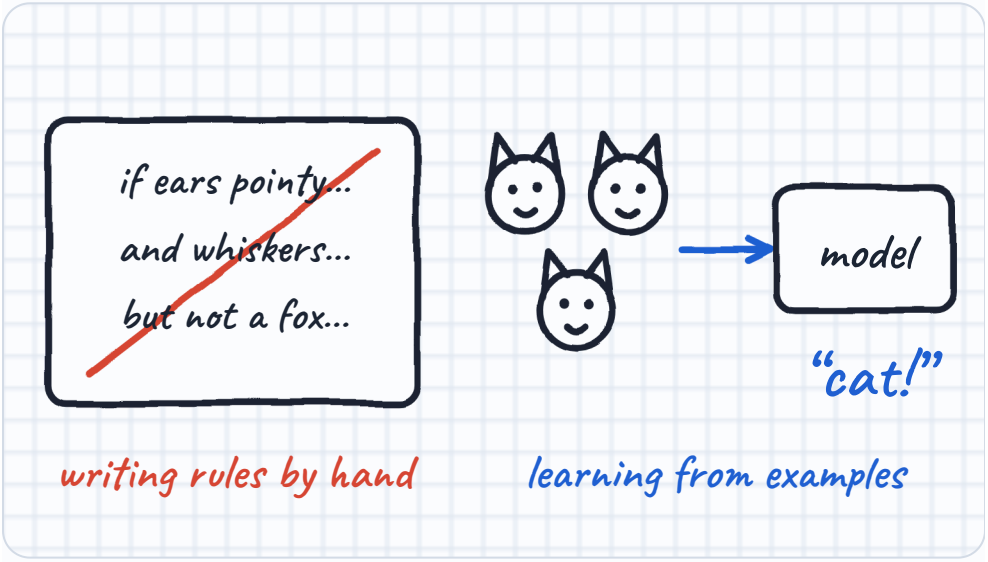
LESSON 1

How machines learn

No magic: examples, a lot of simple maths and millions of small corrections.

CONCEPT

Learning from examples, not rules

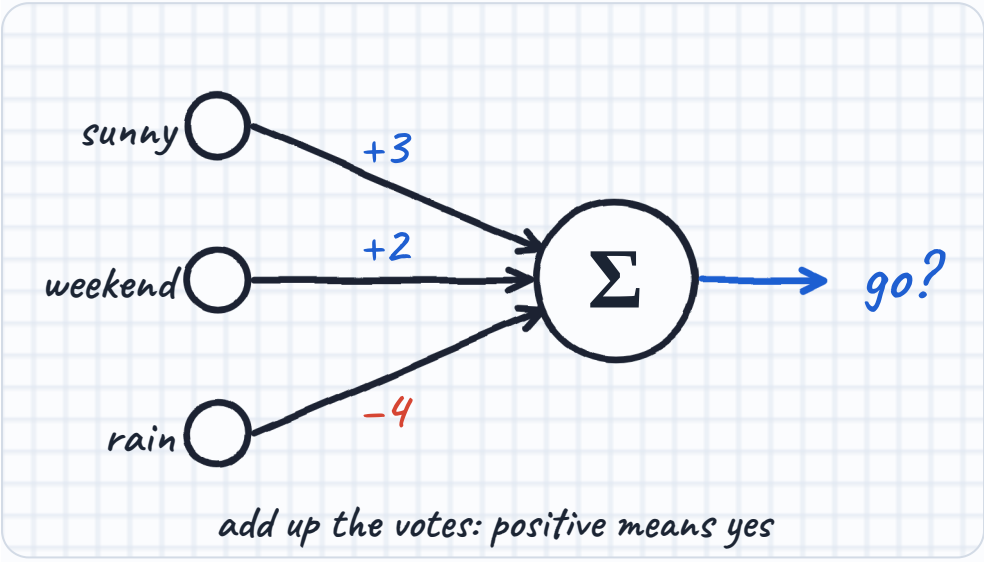


Try writing rules that recognise a cat: pointy ears, whiskers... but foxes have those too, and some cats hide their ears. You'd never finish.

Machine learning flips it around. You show the computer thousands of labelled photos and let it work out the patterns itself. The result is called a model.

CONCEPT

A neuron is a weighted vote

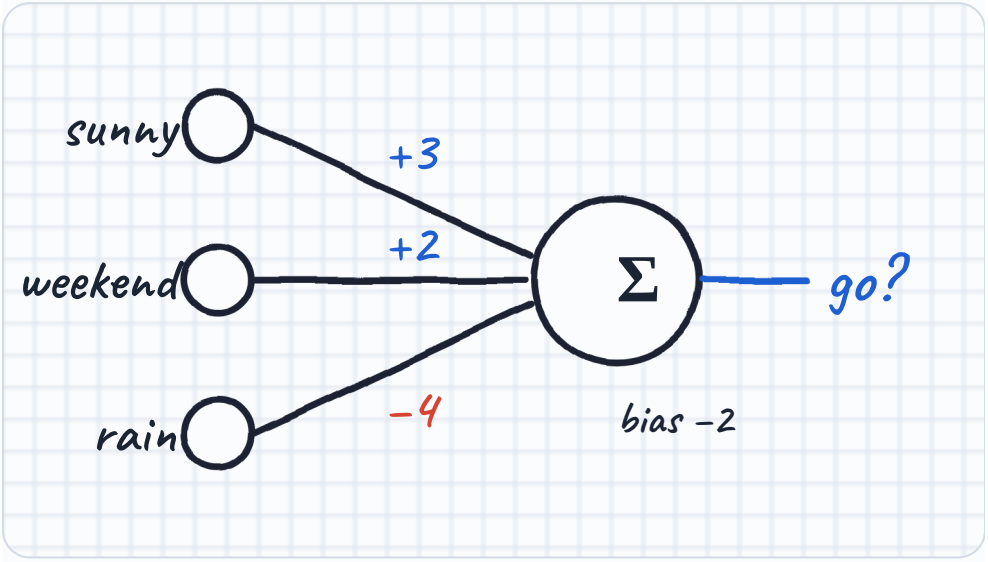


The building block of a neural network is tiny. It takes a few inputs, multiplies each by a weight (how much it matters), adds them up and says yes if the total is high enough.

Should you go to the beach? Sunny counts +3, weekend +2, rain −4. Stack millions of these little votes in layers and you get a network that can recognise faces or write text.

TRY IT

Try it: be the neuron



Switch the inputs on and off. The neuron adds up input × weight, plus a small bias of −2 (it needs a reason to go), and fires when the total is above zero.

CONCEPT

Training means nudging

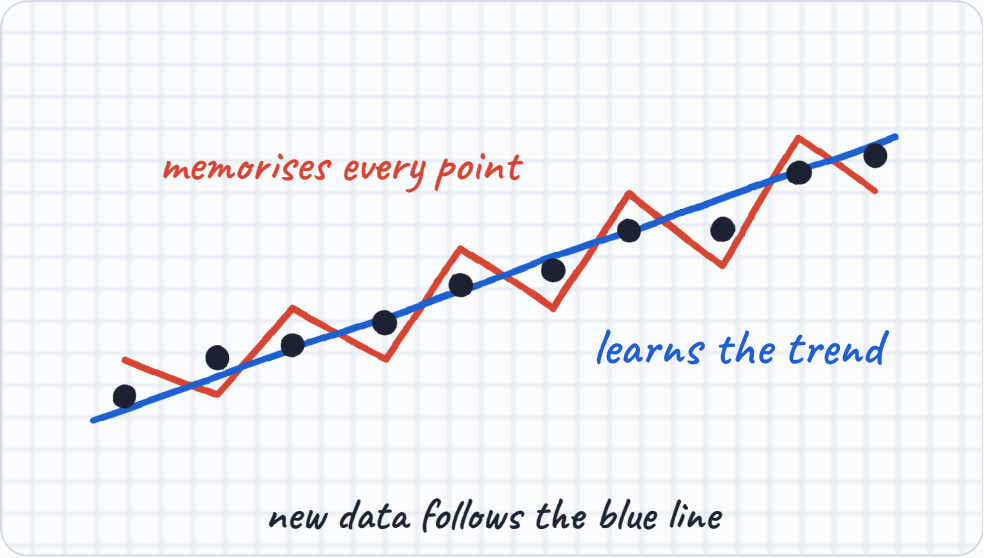


A new network starts with random weights and makes terrible guesses. Training repeats a simple loop: make a guess, measure how wrong it was, then nudge every weight a tiny bit in the direction that reduces the error.

Picture a ball rolling downhill into the valley of lowest error. Large models repeat this billions of times.

FACT

Memorising isn't learning



A model can fit its training examples perfectly and still be useless, like a student who memorised last year's exam answers. This is called overfitting.

The fix is to test on examples the model has never seen. If it does well there too, it has learned the real pattern rather than the noise.

QUIZ

1. What does a machine-learning model learn from?

- Rules a programmer writes
- ✓ **Lots of examples**
- The internet's search results at runtime

It finds patterns in example data. Nobody writes the rules by hand.

2. Sunny (+3) and raining (−4), not a weekend, bias −2. Does the beach neuron fire?

- Yes
- ✓ **No**

3 − 4 − 2 = −3, which is below zero, so no.

3. A model scores 100% on its training data but badly on new data. That's…

- ✓ **Overfitting**
- Perfect training
- A small model

It memorised the examples instead of the general pattern.

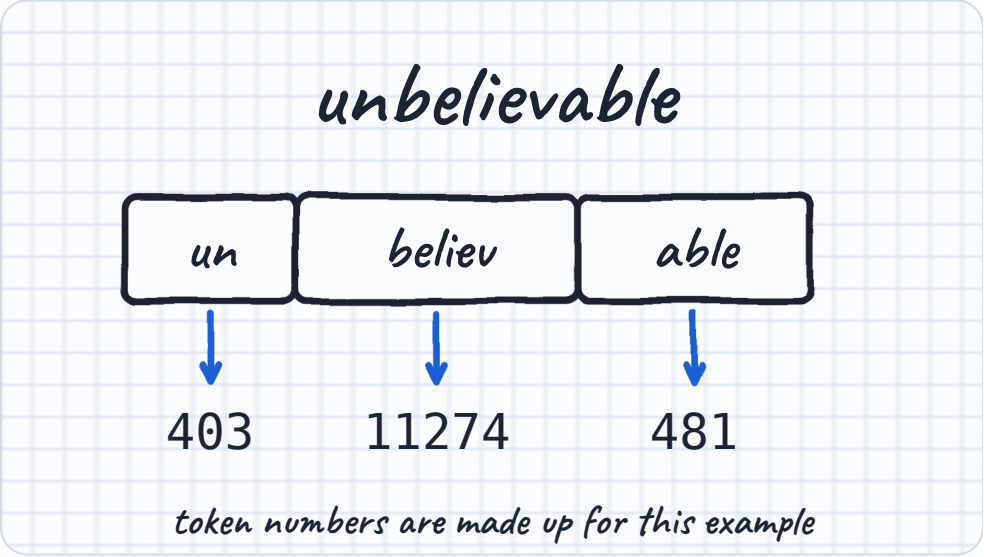
LESSON 2

How chatbots work

What's really going on inside ChatGPT, Claude and friends.

CONCEPT

Text becomes tokens

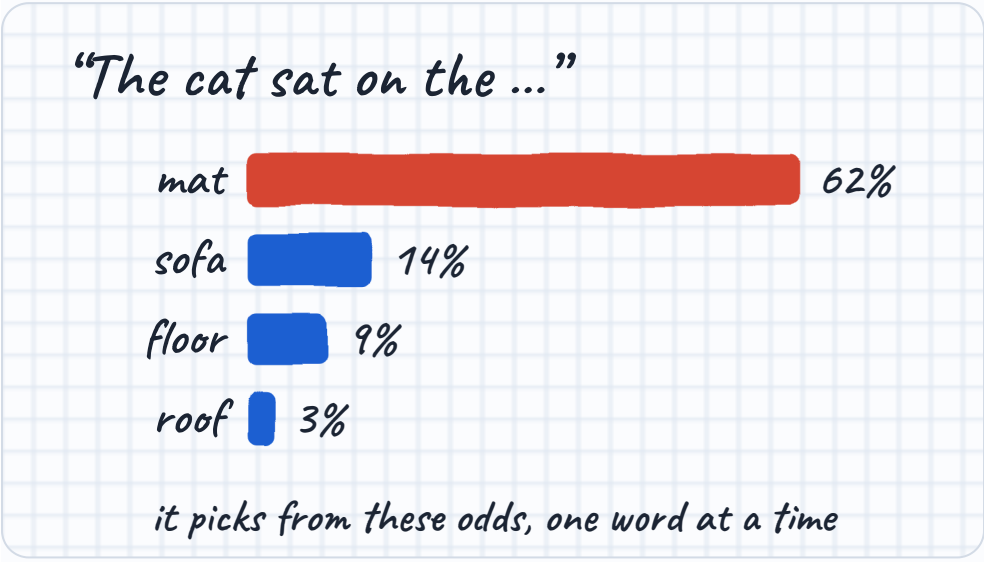


Language models don't read letters or whole words. Text is chopped into tokens: common words stay whole, rarer ones are split into pieces. Each token is just a number to the model.

In English a token averages about ¾ of a word, which is why AI services price and limit usage in tokens.

CONCEPT

It predicts the next word

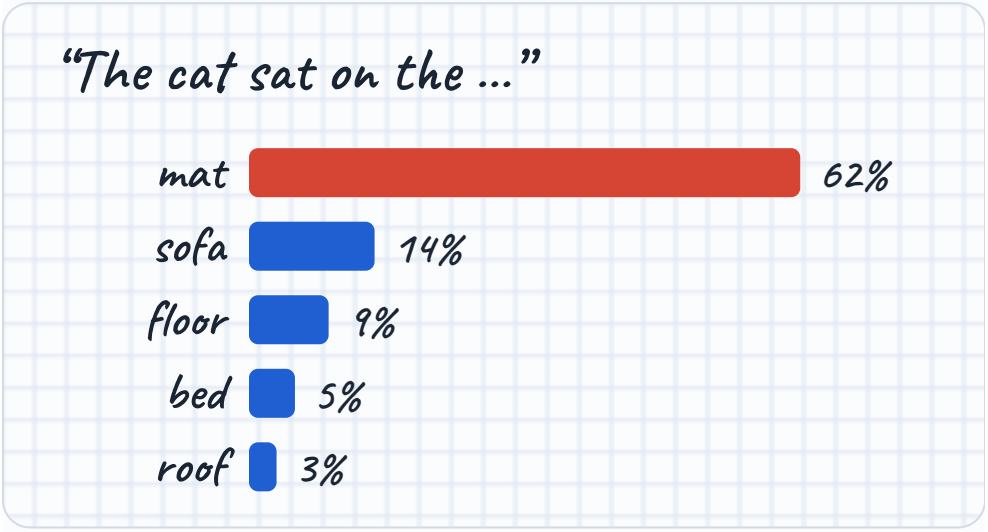


At heart, a language model does one thing: given the text so far, it estimates how likely each possible next token is. It picks one, adds it to the text and repeats.

Trained on a huge amount of writing, that simple trick ends up capturing grammar, facts, styles and a surprising amount of reasoning.

TRY IT

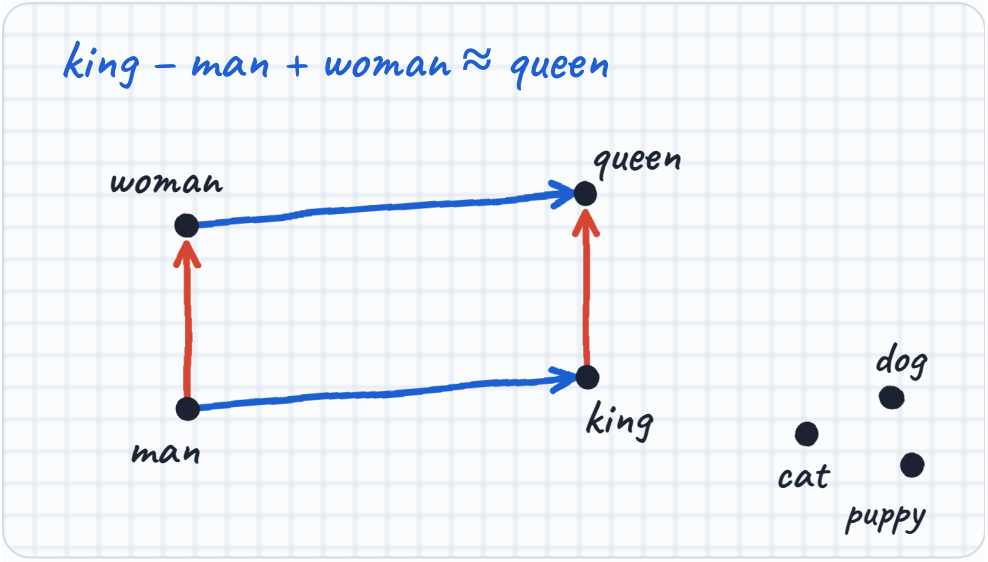
Try it: turn up the temperature



Temperature controls how adventurous the model is. Low temperature almost always picks the top word; high temperature spreads the odds so rarer words get a chance. Press Generate a few times at each setting.

CONCEPT

Meaning as a map

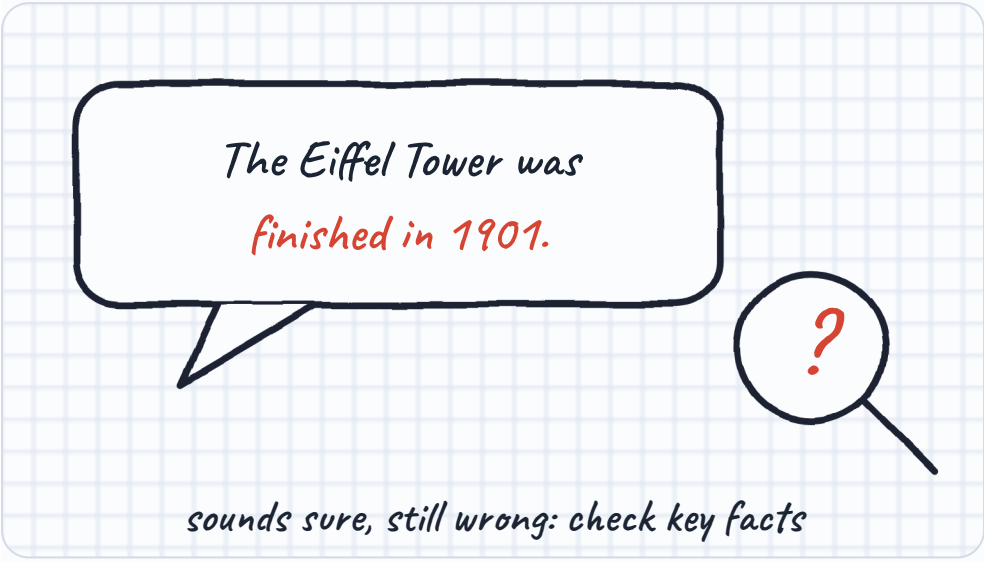


Inside the model, every token becomes a list of numbers called an embedding, like coordinates on a map of meaning. Similar ideas sit close together: cat near dog, Lisbon near Porto.

Directions carry meaning too. The step from 'man' to 'woman' is roughly the same as from 'king' to 'queen'.

FACT

Fluent isn't the same as true



Because a model predicts plausible text, it can produce confident sentences that are simply wrong. People call this hallucination. It's most likely with exact numbers, dates, quotes and niche facts.

Use AI for drafting, explaining and brainstorming, and check anything important against a reliable source. (The Eiffel Tower was finished in 1889.)

QUIZ

1. What does a language model actually do at each step?

- Looks the answer up in a database
- ✔ **Predicts the most likely next token**
- Copies a matching web page

It estimates probabilities for the next token, picks one, and repeats.

2. Roughly how many English words fit in 1,000 tokens?

- About 250
- ✔ **About 750**
- About 4,000

A token averages about ¾ of a word in English.

3. You want consistent, predictable answers. Which temperature?

- ✔ **Low**
- High

Low temperature concentrates the odds on the most likely tokens.

ENGLISH

English

Gramática e vocabulário em inglês, explicados em português.

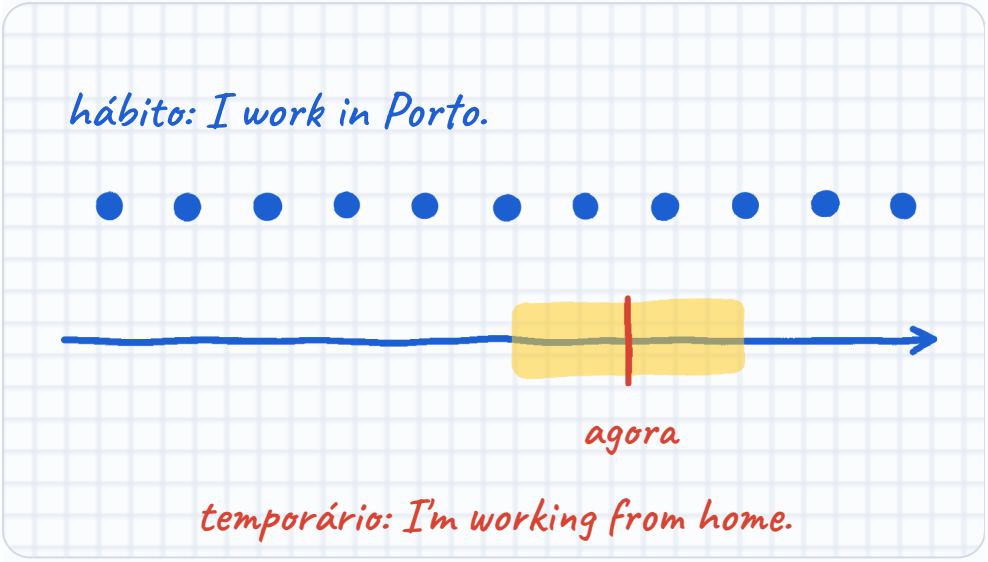
LESSON 1

Tempos verbais sem confusão

Os quatro pares de tempos que mais baralham quem fala português.

REGRA

Present simple ou present continuous?



Usa o ***present simple*** para hábitos, rotinas e verdades gerais: ***I work in Porto.*** (Trabalho no Porto.)

Usa o ***present continuous*** (am/is/are + -ing) para o que está a acontecer agora ou é temporário: ***I'm working from home this week.*** (Esta semana estou a trabalhar em casa.)

Truque: se em português dirias "estou a...", em inglês é quase sempre o continuous.

REGRA

Past simple ou present perfect?



O ***past simple*** fala de um momento terminado e definido: ***I visited London in 2019.***

O ***present perfect*** (have/has + particípio) liga o passado ao presente, sem dizer quando: ***I have visited London twice.*** (Já visitei Londres duas vezes.)

Erro típico: ***I have seen him yesterday*** ×. Com yesterday, last week, in 2019 ou ago, usa sempre o past simple: ***I saw him yesterday*** ✓.

REGRA

For ou since?



for + duração · since + início

Os dois traduzem o nosso "há" ou "desde" quando falamos de algo que ainda continua. Usa ***for*** com quanto tempo dura; usa ***since*** com o momento em que começou. Ambos pedem o present perfect.

WORKED EXAMPLE

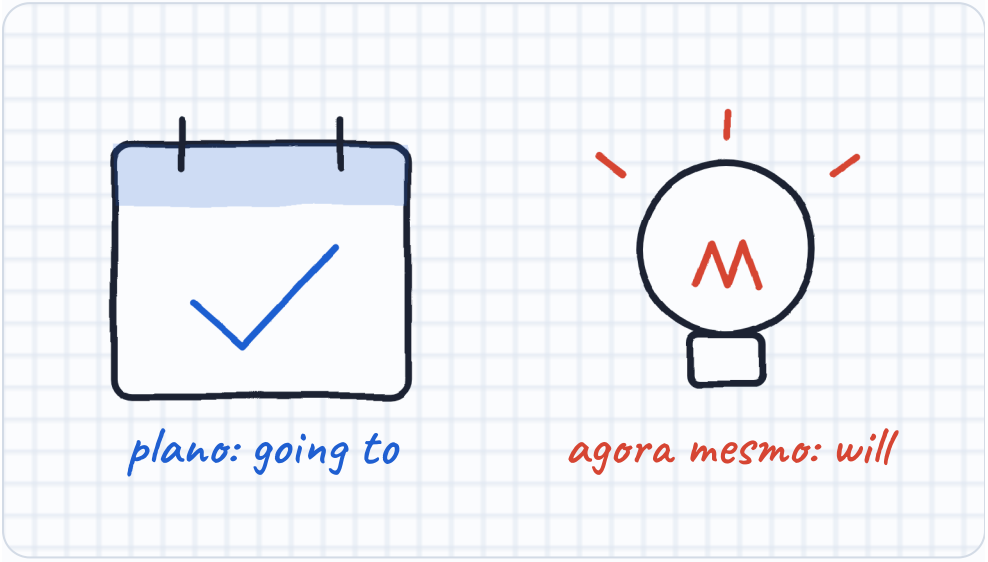
Moro aqui há 10 anos.

I have lived here for 10 years.

= I have lived here since 2016.

REGRA

Will ou going to?



Going to é para planos já decididos ou quando há um sinal à vista: ***I'm going to fly to London on Friday. Look at those clouds, it's going to rain.***

Will é para decisões tomadas no momento, promessas e previsões gerais: ***The phone's ringing. I'll get it! I'll call you later.***

QUIZ

1. I ___ him yesterday.

- have seen
- ✔ **saw**
- see

Com yesterday (momento terminado e definido) usa-se o past simple: I saw him yesterday.

2. She has worked here ___ 2020.

- for
- ✔ **since**
- during

2020 é o ponto de partida, por isso since. Com uma duração seria for (for six years).

3. Look at the sky! It ___ rain.

- will
- ✔ **is going to**
- rains

Há um sinal visível agora (as nuvens), por isso going to.

4. Right now I ___ an email.

- write
- ✔ **am writing**
- wrote

Está a acontecer neste momento: present continuous (am writing).

LESSON 2

Falsos amigos

Palavras que parecem portuguesas mas querem dizer outra coisa.

VOCABULÁRIO

Parece, mas não é

English	parece...	significa
actually	~~atualmente~~	na verdade
pretend	~~pretender~~	fingir
push	~~puxar~~	empurrar
library	~~livraria~~	biblioteca

Os falsos amigos são palavras inglesas parecidas com palavras portuguesas, mas com outro significado. São a fonte de muitos erros de quem fala bem inglês.

Actually não é "atualmente" (isso é ***currently***). ***Pretend*** não é "pretender" (isso é ***intend***). E numa porta, ***push*** quer dizer empurrar, não puxar.

VOCABULÁRIO

Mais quatro armadilhas

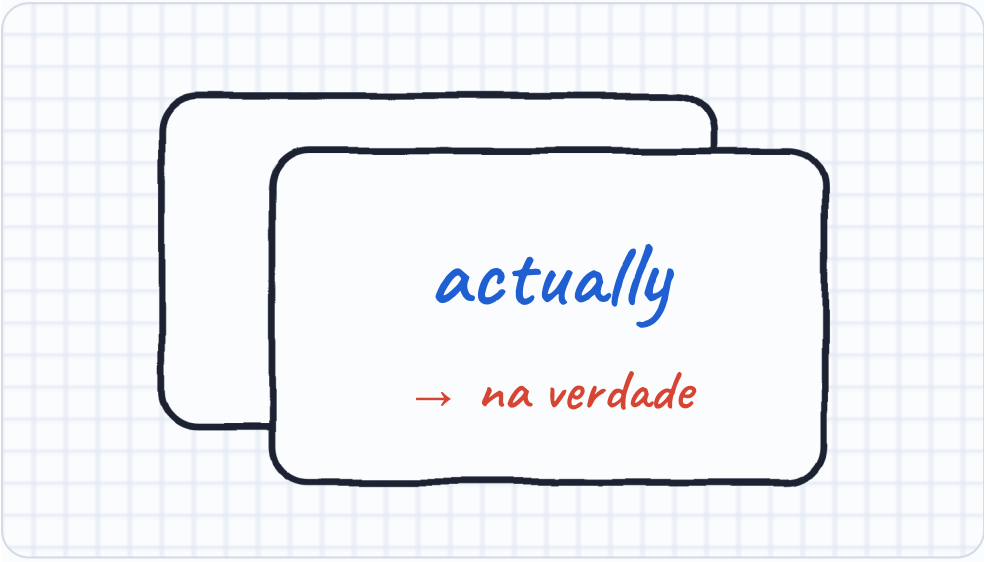
English	parece...	significa
exquisite	~~esquisito~~	requintado
assist	~~assistir~~	ajudar
eventually	~~eventualmente~~	por fim
parents	~~parentes~~	pais

- ***exquisite*** = requintado (esquisito é ***weird*** ou ***picky***)
- ***assist*** = ajudar (assistir a uma reunião é ***attend***; ver um jogo é ***watch***)
- ***eventually*** = por fim, finalmente (eventualmente é ***possibly***)
- ***parents*** = pai e mãe (os parentes são ***relatives***)

Exemplo: ***I'll attend the meeting*** = vou assistir à reunião.

TREINA

Treina: 12 falsos amigos



Lê a palavra em inglês, tenta lembrar-te do significado e toca no cartão para confirmar. Usa Ouvir para treinar a pronúncia.

- **actually** = na verdade, de facto (atualmente = currently)
  Actually, I've never been to Paris.
- **pretend** = fingir (pretender = to intend)
  He pretended to be asleep.
- **push** = empurrar (puxar = pull)
  Push the door to open it.
- **library** = biblioteca (livraria = bookshop)
  I borrowed this book from the library.

- **exquisite** = requintado, delicado (esquisito = weird, picky)
  What an exquisite dinner!
- **assist** = ajudar (assistir a = attend, watch)
  Can I assist you with your bags?
- **eventually** = finalmente, por fim (eventualmente = possibly)
  The flight eventually took off at ten.
- **parents** = pais (mãe e pai) (parentes = relatives)
  My parents live in Viana.
- **realize** = perceber, dar-se conta (realizar = carry out, achieve)
  I didn't realize it was so late.
- **constipated** = com prisão de ventre (constipado = I have a cold)
  I can't come, I have a cold.
- **sympathetic** = compreensivo, solidário (simpático = nice, friendly)
  She was very sympathetic when I was ill.
- **fabric** = tecido (fábrica = factory)
  This fabric is made of cotton.

VOCABULÁRIO

Sensible, sensitive, sympathetic

English	parece...	significa
sensible	~~sensível~~	sensato
sensitive	–	sensível
sympathetic	~~simpático~~	compreensivo
nice, friendly	–	simpático

Três palavras que enganam quase toda a gente:

- ***sensible*** = sensato, com bom senso: ***a sensible decision***
- ***sensitive*** = sensível: ***sensitive skin***, ***a sensitive topic***
- ***sympathetic*** = compreensivo, solidário. Para "simpático" diz ***nice*** ou ***friendly***.

QUIZ

1. "Eventually" significa...

- eventualmente
- ✓ **por fim, finalmente**
- frequentemente

Eventually = por fim. Para "eventualmente" diz possibly ou perhaps.

2. Que palavra quer dizer "atualmente"?

- actually
- ✓ **currently**
- eventually

Currently = atualmente. Actually = na verdade.

3. Uma porta diz PULL. Deves...

- empurrar
- ✓ **puxar**

Pull = puxar. Push = empurrar.

4. "My manager is very sympathetic" quer dizer que é...

- muito simpático
- ✓ **compreensivo, solidário**

- muito sensível

Sympathetic = compreensivo. Simpático diz-se nice ou friendly.

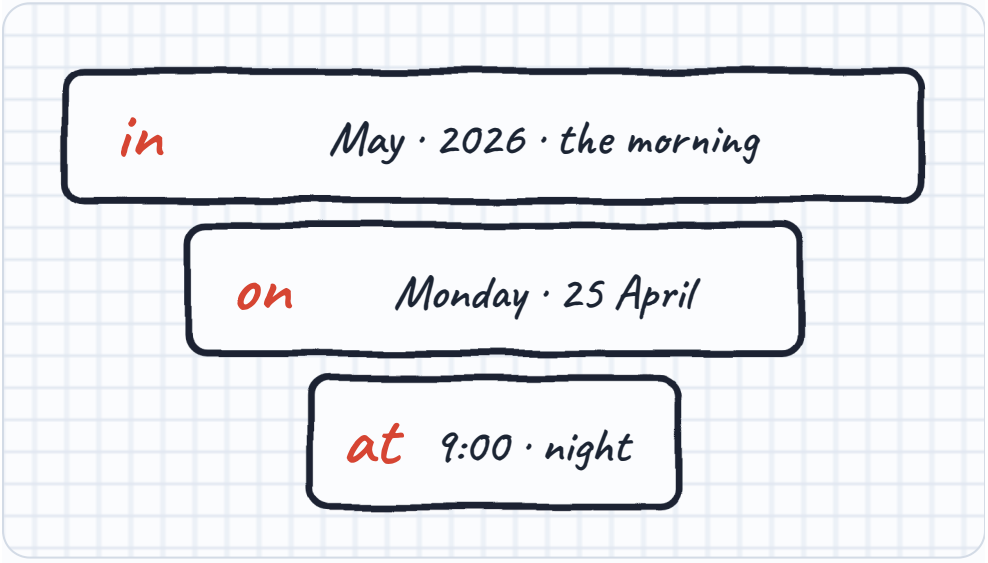
LESSON 3

Preposições e verbos do dia a dia

in, on, at; make ou do; phrasal verbs e os erros mais comuns.

REGRA

In, on, at no tempo

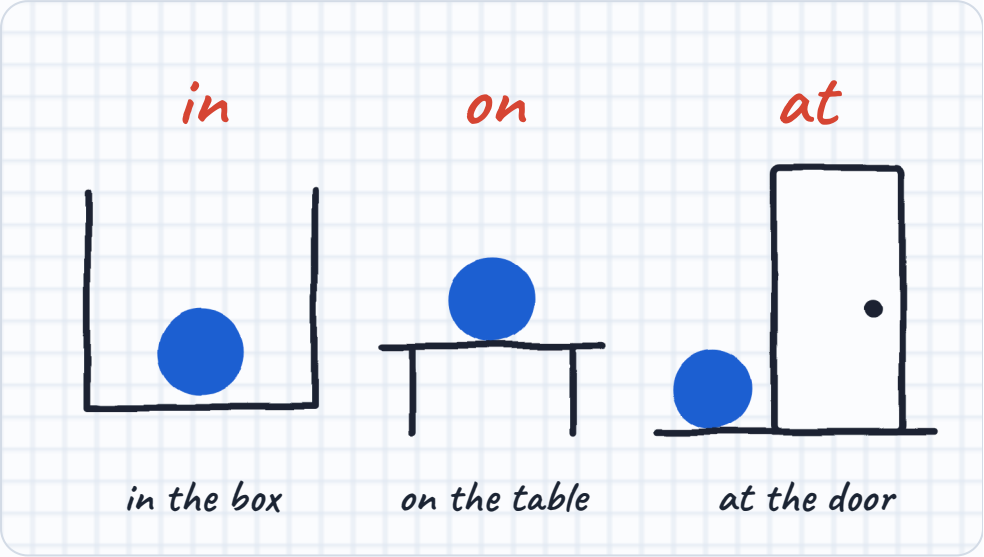


Pensa num funil, do mais largo para o mais preciso:

- ***in*** para períodos longos: meses, anos, estações e partes do dia (***in May***, ***in 2026***, ***in the morning***)
- ***on*** para dias e datas: ***on Monday***, ***on 25 April***
- ***at*** para horas e momentos exatos: ***at 9:00***, ***at night***, ***at the weekend*** (inglês britânico)

REGRA

In, on, at no espaço



- *in* = dentro de um espaço fechado ou de uma área: *in the box*, *in the car*, *in Lisbon*
- *on* = em cima de uma superfície: *on the table*, *on the wall*, *on the second floor*
- *at* = num ponto ou local concreto: *at the door*, *at the airport*, *at work*

VOCABULÁRIO

Make ou do?

MAKE	DO
criar · produzir	tarefas · atividades
a decision	homework
a mistake	the dishes
money	business
a cake	your best

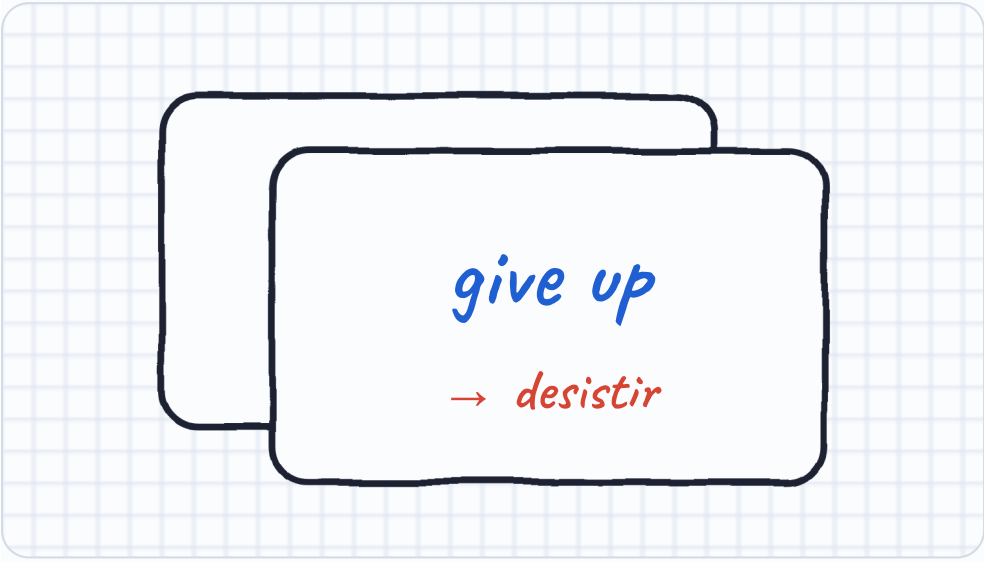
Os dois traduzem "fazer".

- ***make*** = criar ou produzir alguma coisa: ***make a decision***, ***make a mistake***, ***make money***, ***make a cake***
- ***do*** = tarefas, atividades e trabalho: ***do homework***, ***do the dishes***, ***do business***, ***do your best***

Não há uma regra perfeita: o melhor é aprender as combinações.

TREINA

Treina: 10 phrasal verbs



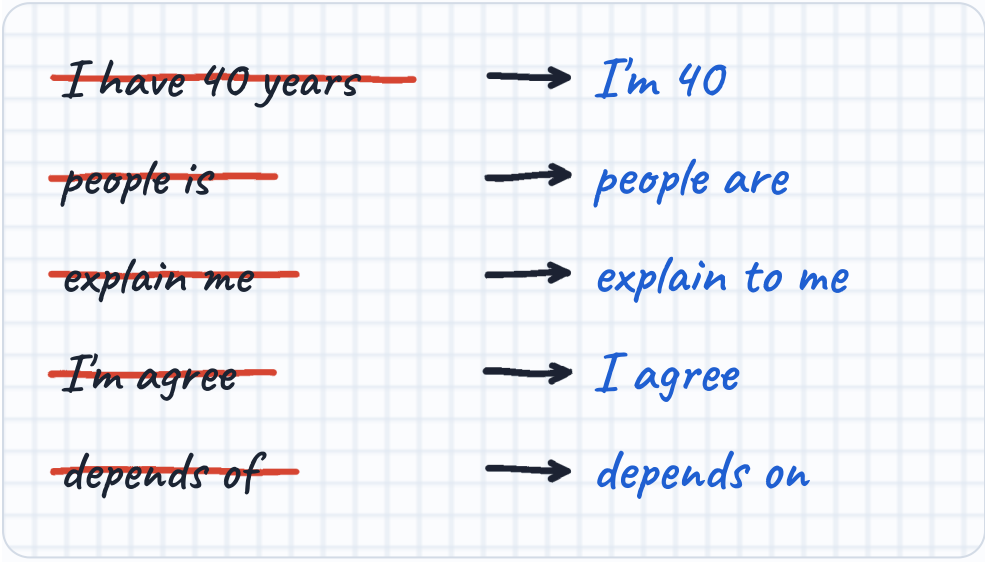
Phrasal verbs são verbos com uma partícula (up, out, after...) que muda o significado. São muito usados no dia a dia e nas reuniões. Toca no cartão para ver a tradução.

- **look up** = procurar (num dicionário, na internet)
  Look it up online.
- **look after** = tomar conta de, cuidar de
  Can you look after the kids tonight?
- **look forward to** = estar ansioso por (coisa boa) (segue-se -ing)
  I'm looking forward to seeing you.

- **give up** = desistir; deixar de
  Don't give up now!
- **run out of** = ficar sem
  We've run out of coffee.
- **figure out** = perceber, descobrir como
  I can't figure out this app.
- **find out** = descobrir (uma informação)
  Let me find out when the train leaves.
- **carry on** = continuar
  Carry on, I'm listening.
- **turn down** = recusar; baixar (o volume)
  She turned down the job offer.
- **set up** = montar, configurar
  I set up a new server at home.

ERROS COMUNS

Cinco erros típicos de portugueses



Traduzir à letra do português leva a estes erros:

- idade usa o verbo ***be***: ***I'm 40*** (ou ***I'm 40 years old***)
- ***people*** é plural: ***people are***
- ***explain*** pede ***to***: ***explain to me***
- ***agree*** é um verbo, sem ***am***: ***I agree***
- ***depend*** usa ***on***: ***it depends on the weather***

QUIZ

1. The meeting is ___ Monday ___ 10:00.

- in / at
- ✓ **on / at**
- at / on

Dias levam on; horas levam at: on Monday at 10:00.

2. We need to ___ a decision today.

- do
- ✓ **make**

Decisões "fazem-se" com make: make a decision.

3. We've run ___ coffee.

- ✓ **out of**
- off
- away

Run out of = ficar sem.

4. Qual está correto?

- I have 40 years.
- ✓ **I'm 40.**

- I'm having 40 years.

Em inglês a idade usa o verbo be: I'm 40 ou I'm 40 years old.

ELECTRICITY

Electricity

From Ohm's law to the fuse board in your hallway.

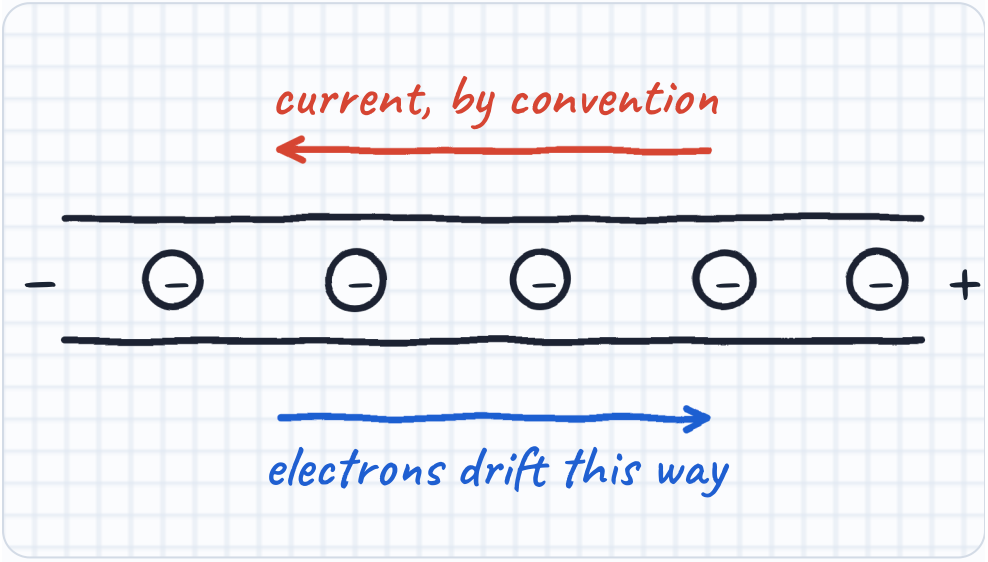
LESSON 1

Voltage, current & resistance

The three quantities behind every circuit, and the one law that ties them together.

CONCEPT

Electricity is moving charge

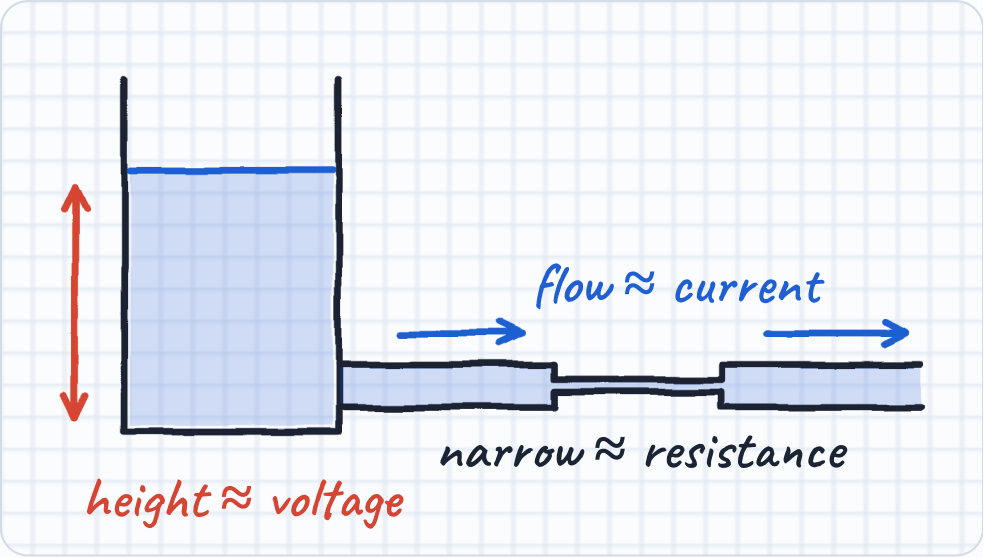


A current is a flow of electric charge. In a metal wire it's electrons nudging each other along. Each electron drifts slowly, but the push travels through the wire almost instantly, like water already filling a hose.

By convention, current is drawn flowing from + to −, the opposite way to the electrons. Every circuit diagram uses this convention.

CONCEPT

The water analogy



Picture a tank feeding a pipe.

- Voltage is the pressure, set by how high the water stands.
- Current is how much water flows per second.
- Resistance is a narrow section of pipe that limits the flow.

More pressure pushes more flow; a narrower pipe allows less. That's the whole idea behind Ohm's law.

FORMULA

Ohm's law



$V = I \times R$

$I = V \div R$

$R = V \div I$

V = I × R

Voltage (volts, V) equals current (amps, A) times resistance (ohms, Ω). Cover the quantity you want in the triangle and the other two tell you how to get it.

WORKED EXAMPLE

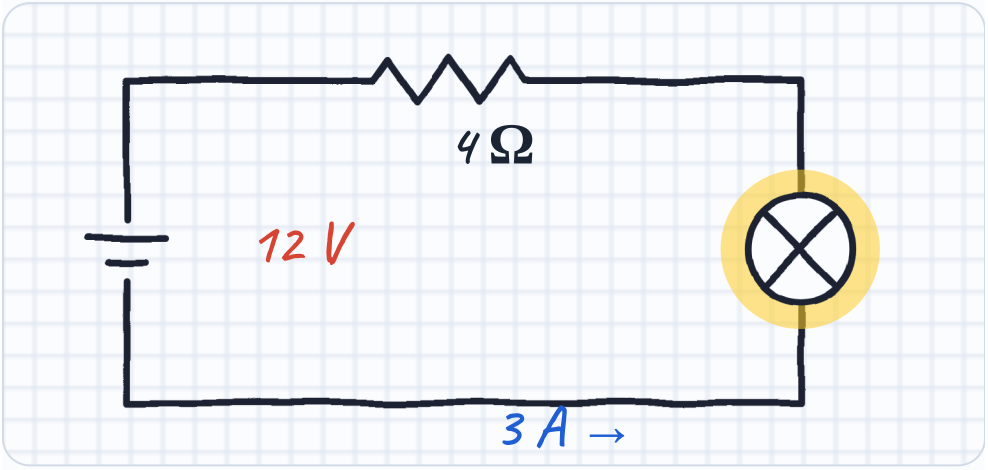
12 V battery across a 4 Ω heater

$I = V \div R = 12 \div 4$

$= 3\ A$

TRY IT

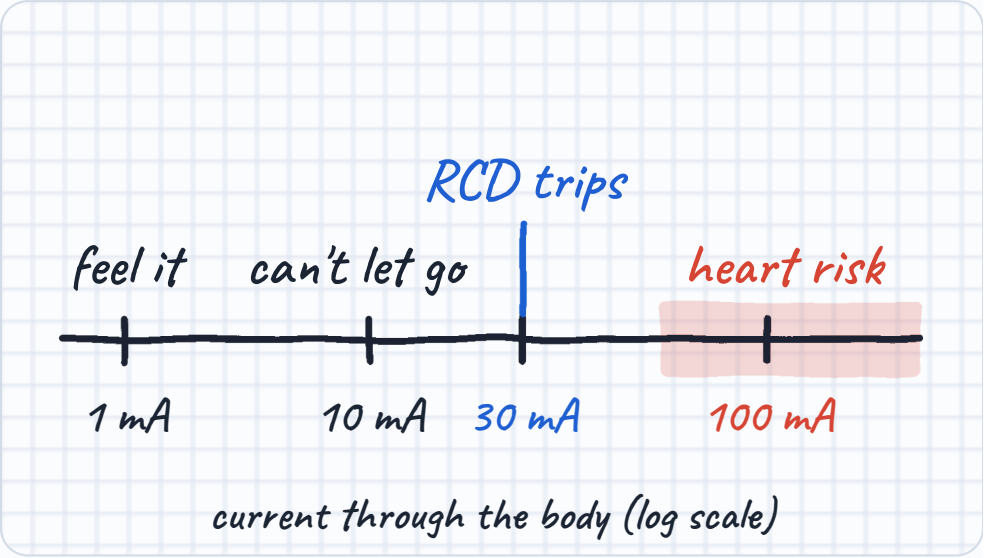
Try it: dim the bulb



Change the voltage and the resistance. Watch the current and the power the bulb turns into light and heat.

FACT

It's the current that hurts



Your body is a resistor too. What matters for safety is how much current passes through it, and which way.

Around 1 mA you feel a tingle. Near 10 mA muscles can lock and you may not be able to let go. From roughly 50–100 mA across the chest, the heart can go into fibrillation.

That's why home RCDs trip at 30 mA: well before the dangerous zone.

QUIZ

1. A 9 V battery drives current through a 450 Ω resistor. What is the current?

- ✔ **20 mA**
- 2 A
- 4 mA
- 50 mA

I = V ÷ R = 9 ÷ 450 = 0.02 A, which is 20 mA.

2. In the water analogy, what plays the role of voltage?

- How fast the water flows
- ✔ **The water pressure (tank height)**
- The width of the pipe

Voltage is the push, like pressure. Flow is current; pipe narrowness is resistance.

3. You double the resistance and keep the voltage the same. The current…

- doubles
- stays the same
- ✔ **halves**

I = V ÷ R. Twice the R with the same V gives half the current.

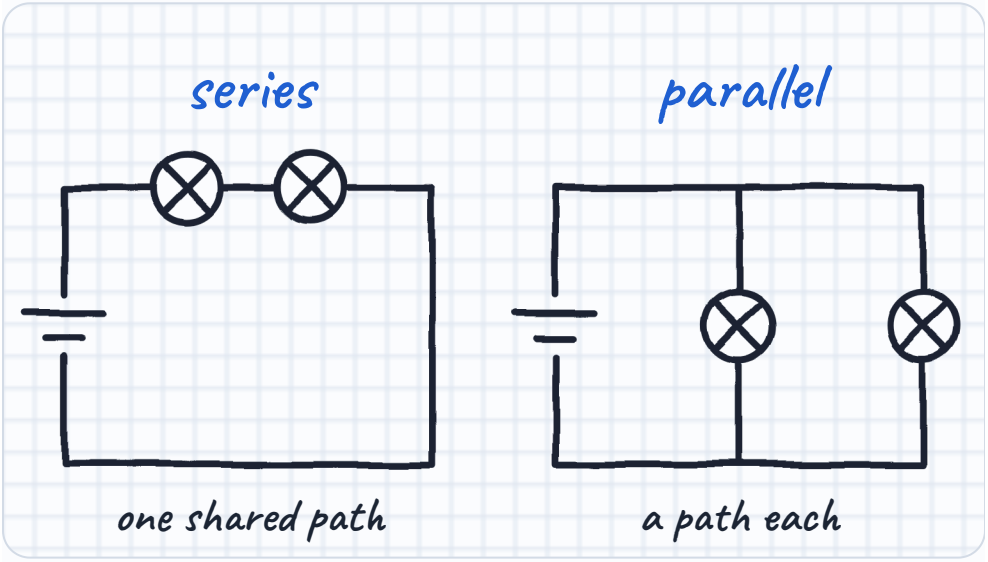
LESSON 2

Circuits at home

Series vs parallel, power, and what the breakers in your board actually do.

CONCEPT

Series vs. parallel

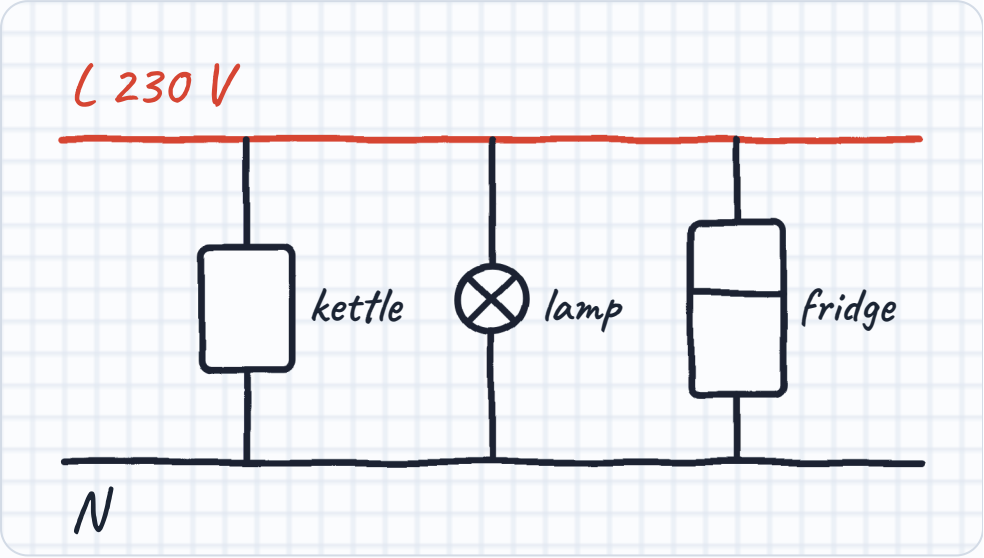


In series, components share a single loop. The same current flows through each and the voltage is split between them. One break stops everything, like old Christmas lights.

In parallel, each component has its own path to the supply. Each gets the full voltage and can be switched on or off independently.

CONCEPT

Your house is wired in parallel



Every socket and light hangs between the live (L) and neutral (N) conductors. That's why each appliance gets the full 230 V, and why switching off the kettle doesn't turn off the fridge.

The more you plug in, the more total current flows in the live wire. That total is what breakers watch.

FORMULA

Power



P = V × I

Power (watts) is how fast energy is used. Multiply the voltage by the current. Rearranged, I = P ÷ V tells you how much current an appliance draws.

WORKED EXAMPLE

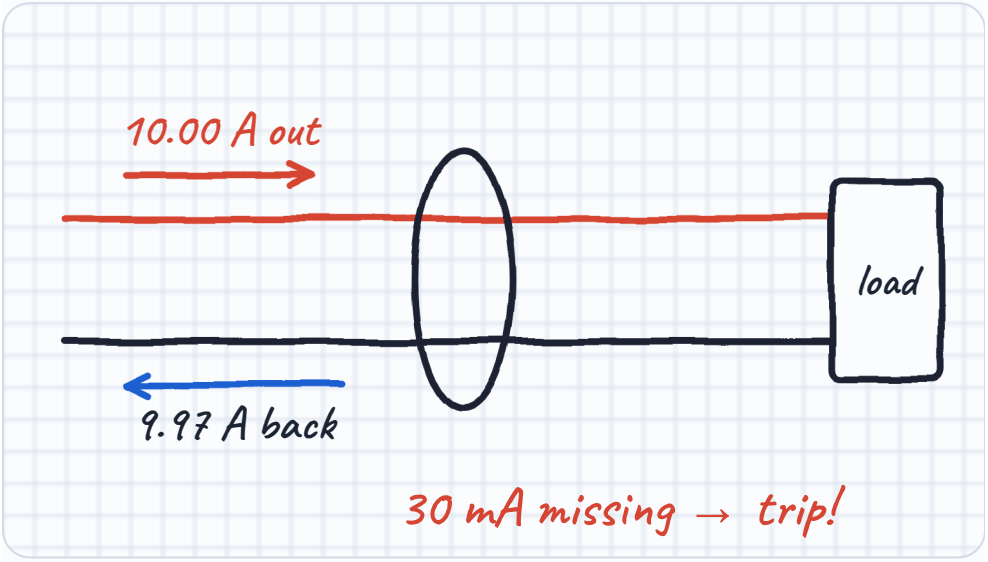
2300 W kettle on 230 V

I = 2300 ÷ 230

= 10 A

CONCEPT

Breakers vs. RCDs

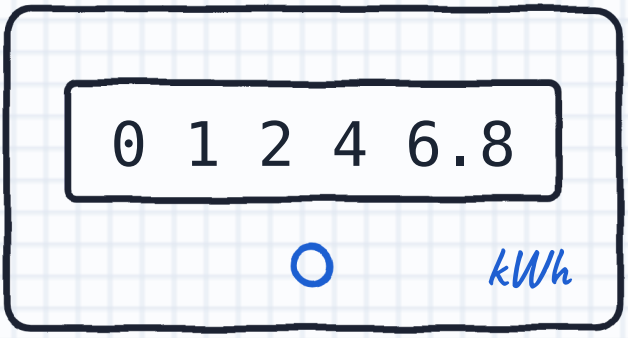


A breaker (MCB) protects the cable. If too much current flows, the wire would overheat, so it trips. A 16 A breaker on a socket circuit is typical.

An RCD protects people. It compares the current going out on live with the current coming back on neutral. If about 30 mA is missing, it has leaked somewhere, maybe through a person, and it trips in a fraction of a second.

WORKED EXAMPLE

Reading your bill: kWh



energy = power × time

You pay for energy, not power. Energy = power × time, measured in kilowatt-hours (kWh). One kWh is 1 kW running for one hour.

WORKED EXAMPLE

2 kW heater for 3 hours

2 kW × 3 h = 6 kWh

at €0.20 per kWh → €1.20

QUIZ

1. Why does each socket in a house get the full 230 V?

- They're wired in series
- ✓ **They're wired in parallel**
- Each socket has its own transformer

In parallel, every load sits directly between live and neutral, so each sees the full supply voltage.

2. A 1150 W heater runs on 230 V. How much current does it draw?

- 0.2 A
- ✓ **5 A**
- 50 A
- 23 A

I = P ÷ V = 1150 ÷ 230 = 5 A.

3. What makes an RCD trip?

- Total current above its rating
- ✓ **Current going out on live doesn't all return on neutral**
- Voltage drops below 200 V

An RCD measures the imbalance between live and neutral. Overcurrent is the breaker's job.

4. A 1.5 kW oven runs for 4 hours. How much energy is that?

- ✔ **6 kWh**
- 1.5 kWh
- 375 Wh

1.5 kW × 4 h = 6 kWh.

ELECTRONICS

Electronics

Components and signals for tinkering with microcontrollers.

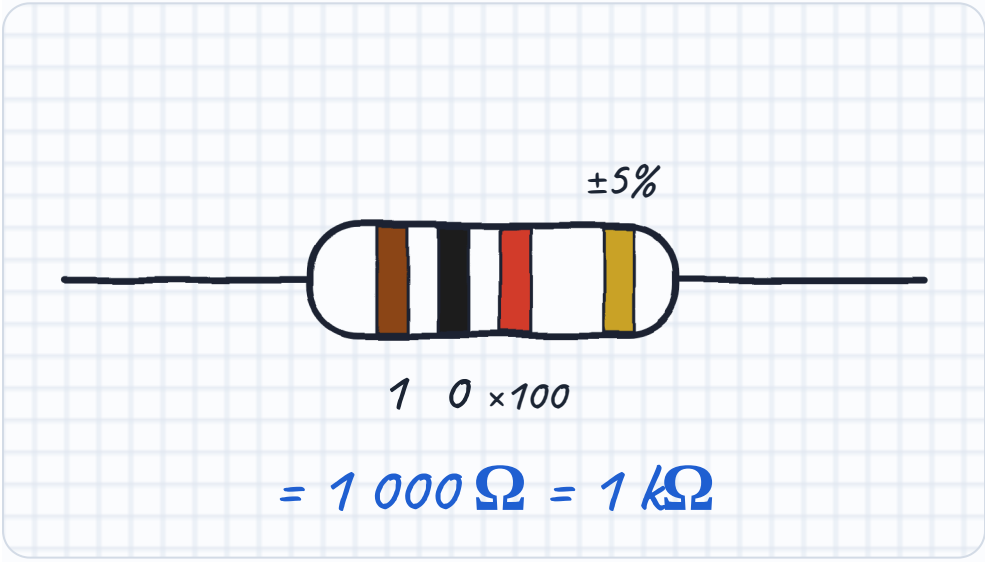
LESSON 1

Components you meet first

Resistors, LEDs, capacitors and the voltage divider: the parts in every maker kit.

CONCEPT

Reading a resistor

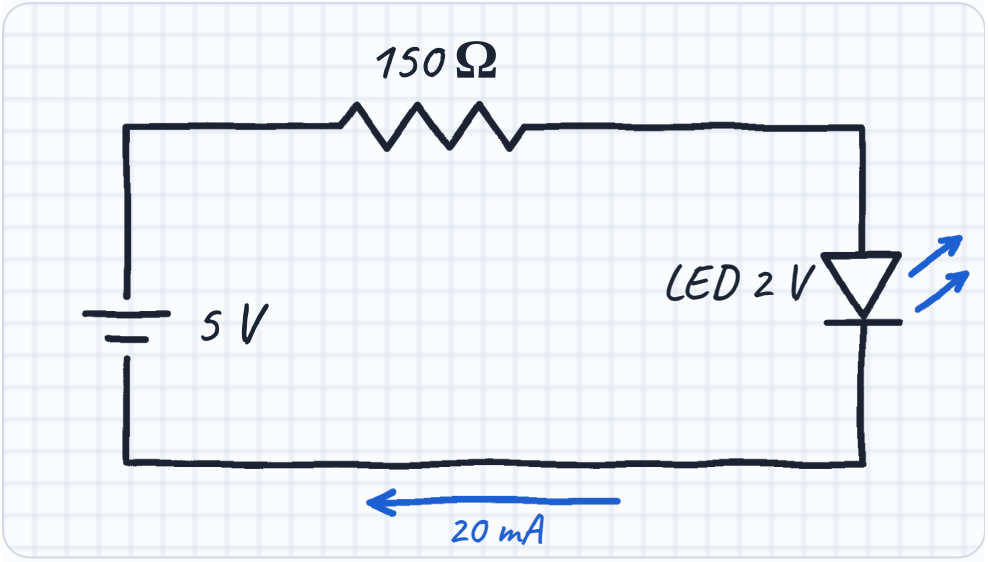


Resistors are labelled with coloured bands. The first two are digits, the third is a multiplier, and a gold or silver band at the end gives the tolerance.

Black 0, brown 1, red 2, orange 3, yellow 4, green 5, blue 6, violet 7, grey 8, white 9.

FORMULA

Every LED needs a resistor



R = (Vs – Vf) ÷ I

An LED drops a fixed forward voltage (Vf, about 2 V for red, 3 V for white or blue) and will draw as much current as it can get, then burn out. A series resistor takes up the leftover voltage and sets the current.

WORKED EXAMPLE

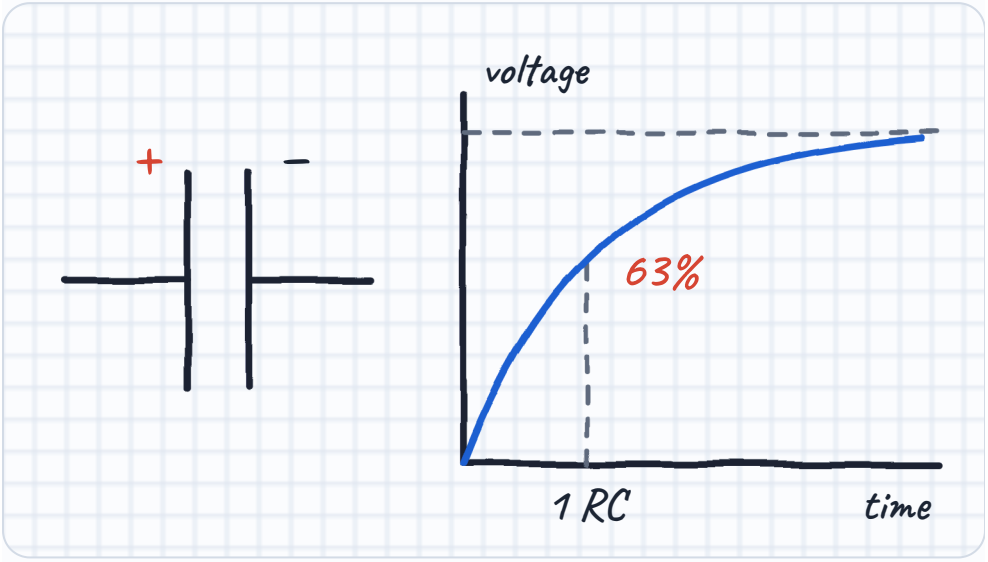
5 V supply, red LED (2 V), 20 mA

$R = (5 - 2) \div 0.020$

$= 150\ \Omega$

CONCEPT

Capacitors store charge

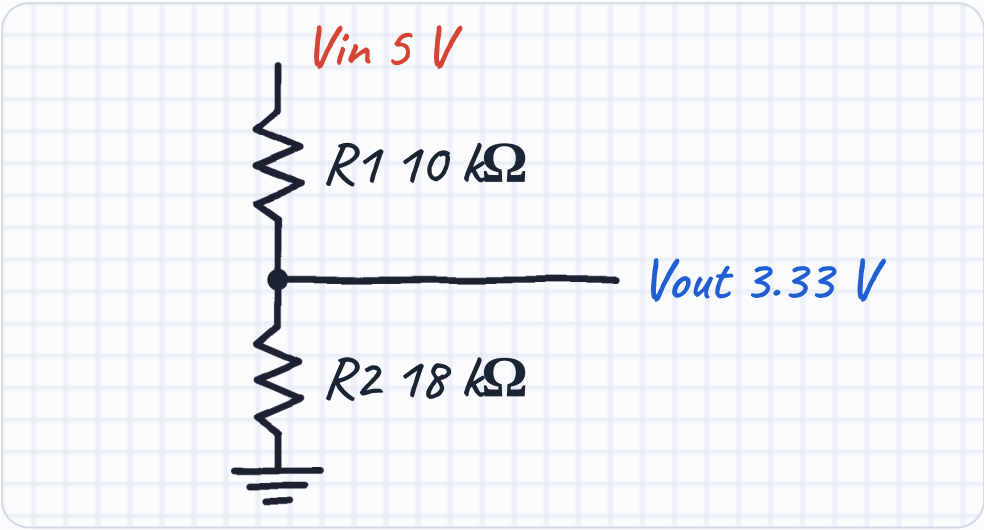


A capacitor is two plates separated by an insulator. Connect it to a voltage and charge piles up on the plates until it matches the supply. It charges quickly at first, then slows down.

In practice they smooth out ripples on a power rail and supply short bursts of current, which is why you see one next to almost every chip.

TRY IT

Try it: the voltage divider



Vout = Vin × R2 ÷ (R1 + R2)

Two resistors in series split a voltage in proportion to their values. It's the standard way to bring a 5 V or 12 V signal down to the 3.3 V an ESP32 pin can take.

QUIZ

1. A white LED (Vf 3 V) runs from 12 V at 10 mA. Which resistor?

- 90 Ω
- 300 Ω
- ✓ **900 Ω**
- 1.2 kΩ

R = (12 − 3) ÷ 0.010 = 900 Ω. You'd pick the next standard value up, 1 kΩ.

2. Bands brown, black, orange. What's the value?

- 10 Ω
- 1 kΩ
- ✓ **10 kΩ**
- 103 Ω

Brown 1, black 0, orange ×1000 → 10 × 1000 = 10 kΩ.

3. In a divider with R1 = R2 and Vin = 10 V, Vout is…

- 10 V
- ✓ **5 V**
- 2.5 V

Equal resistors split the voltage in half.

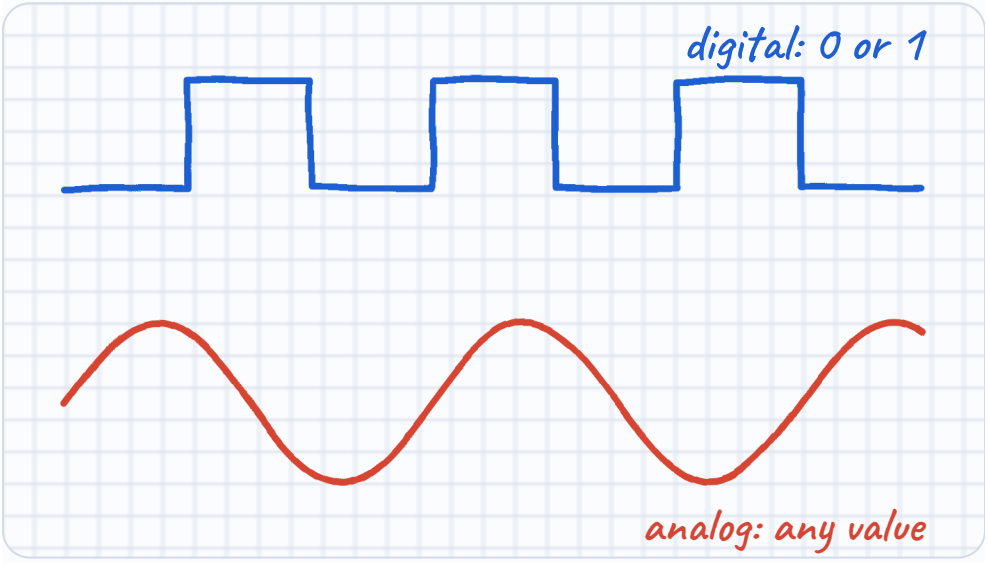
LESSON 2

Signals & microcontrollers

How a chip like the ESP32 reads and controls the real world.

CONCEPT

Digital vs. analog

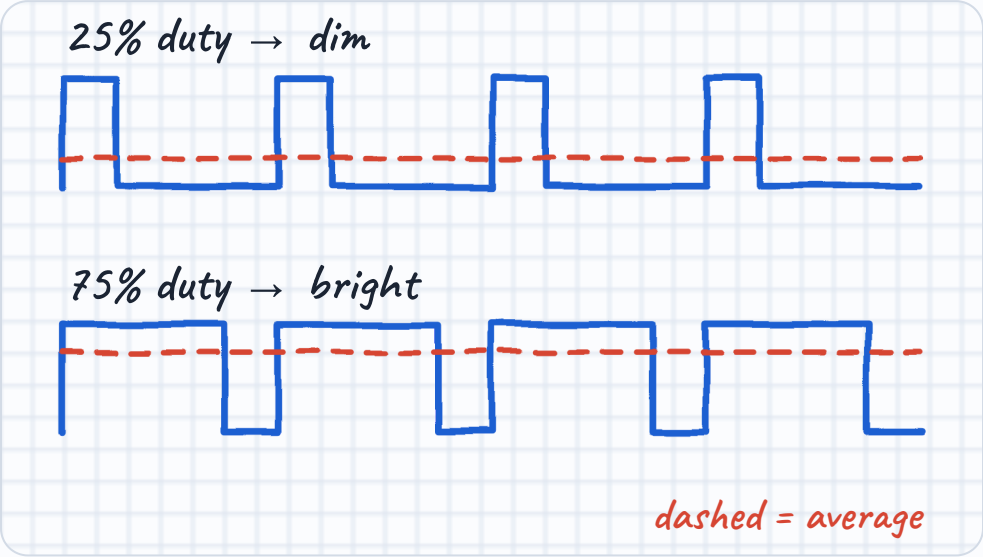


A digital signal has two states: low (0) or high (1, often 3.3 V). An analog signal can be any value in a range, like a temperature sensor's output.

Microcontrollers are digital at heart. To deal with analog they use an ADC to read and PWM to fake an output.

CONCEPT

PWM: dimming by blinking

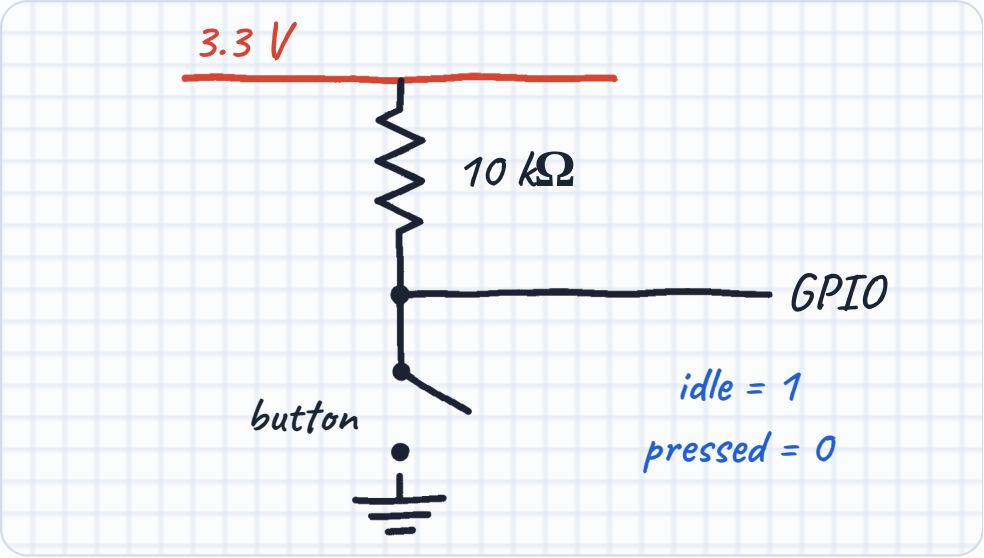


Pulse-width modulation switches a pin fully on and off very fast. The fraction of time it's on is the duty cycle. An LED or motor responds to the average: 25% duty on a 12 V supply behaves roughly like 3 V.

It's efficient because the switch is always fully on or fully off, so little energy is wasted as heat.

CONCEPT

Pull-up resistors

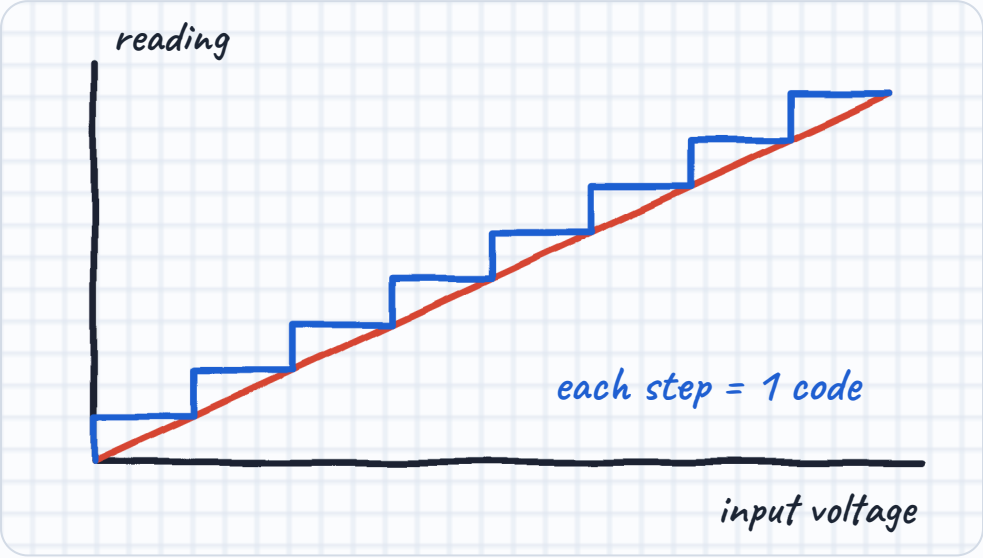


An input pin connected to nothing 'floats' and picks up noise, randomly reading 0 or 1. A pull-up resistor ties it gently to 3.3 V so it reads 1 by default. Press the button and the pin connects to ground, reading 0.

Most microcontrollers have internal pull-ups you can switch on in code.

FORMULA

ADC resolution



$step = Vref \div 2^n$

An analog-to-digital converter turns a voltage into a number. An n-bit ADC has 2^n steps. The ESP32's 12-bit ADC gives 4096 steps over about 3.3 V.

Its readings are less accurate near 0 V and near the top of the range, so keep signals in the middle where you can.

WORKED EXAMPLE

12-bit ADC, 3.3 V reference

3.3 ÷ 4096

≈ 0.8 mV per step

QUIZ

1. A 12 V motor gets PWM at 50% duty. It behaves roughly as if it had...

- 12 V
- ✓ **6 V**
- 0.5 V
- 24 V

The average of 12 V on half the time is 6 V.

2. What does a 'floating' input do?

- Always reads 1
- Always reads 0
- ✓ **Reads randomly because it picks up noise**

With nothing defining its level, the pin picks up stray charge. A pull-up or pull-down fixes that.

3. How many steps does a 10-bit ADC have?

- 10
- 100
- ✓ **1024**
- 4096

$2^{10} = 1024$.

ENGINEERING

Engineering

Forces, machines and the shapes that keep structures standing.

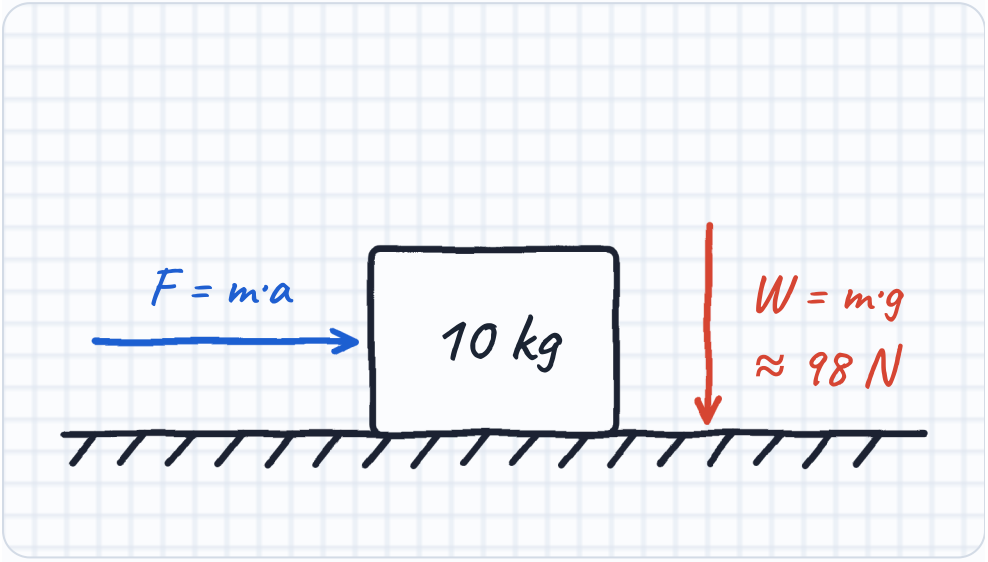
LESSON 1

Forces, levers & gears

Why a long spanner works, and how machines trade force for distance.

FORMULA

Force, mass and weight



F = m × a

A force makes a mass accelerate. Weight is just the force of gravity on a mass: W = m × g, with g ≈ 9.81 m/s² on Earth. Mass is measured in kg; force and weight in newtons (N).

WORKED EXAMPLE

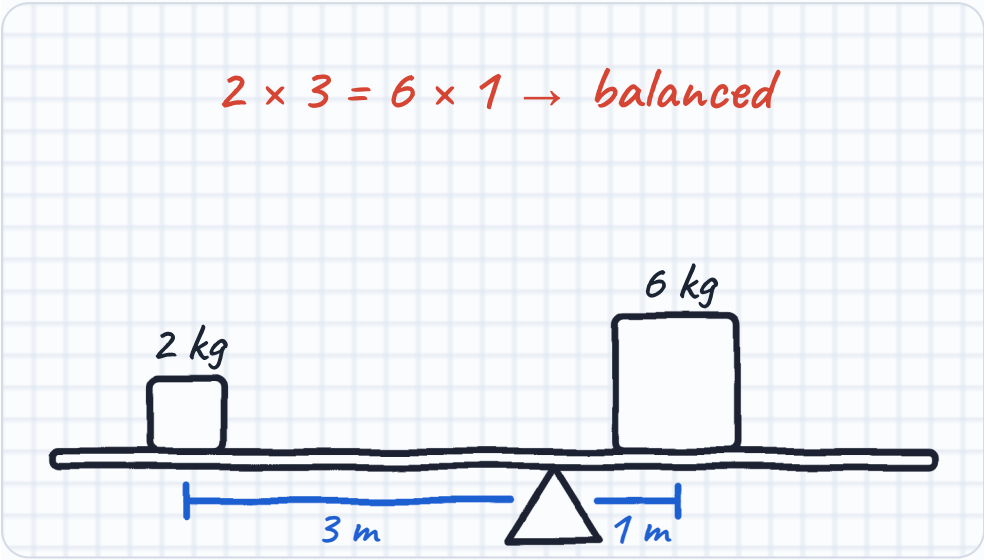
A 10 kg box

W = 10 × 9.81

≈ 98 N

CONCEPT

Levers multiply force



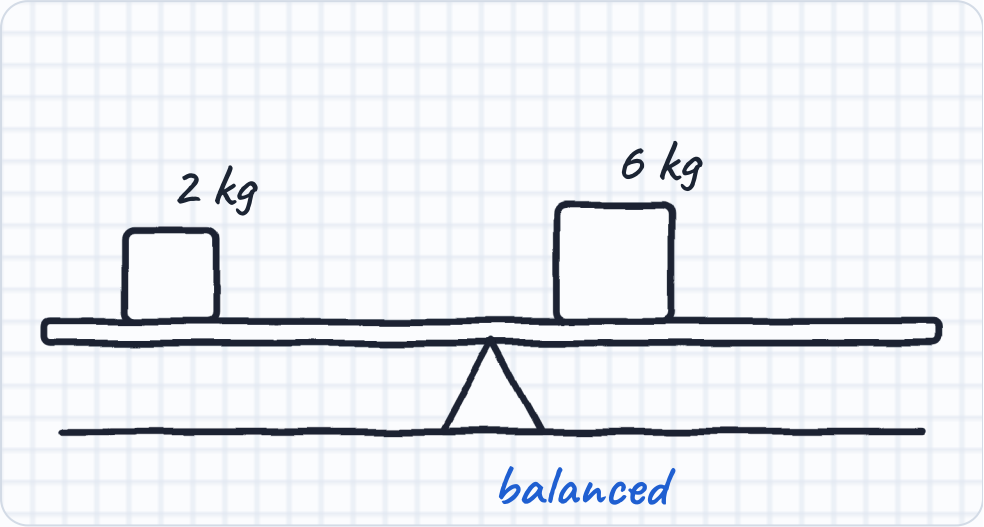
torque = F × d

Turning effect, or torque, is force × distance from the pivot. A small force far from the pivot balances a large force close to it.

That's why a long spanner loosens a stuck bolt, and why door handles sit far from the hinges.

TRY IT

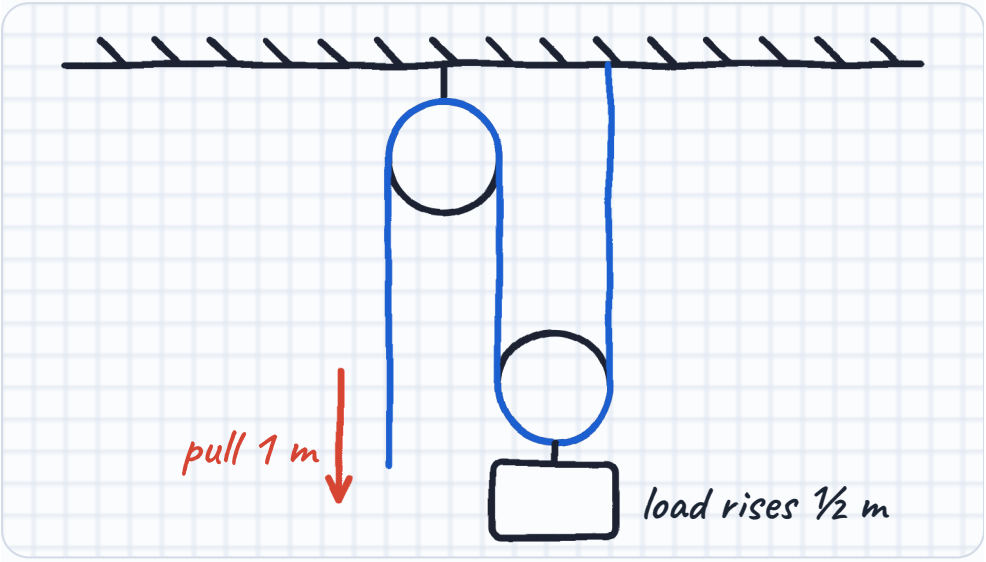
Try it: balance the seesaw



The right side holds 6 kg at 1 m from the pivot. Change the mass and distance on the left until the beam levels out.

CONCEPT

Pulleys trade distance for force

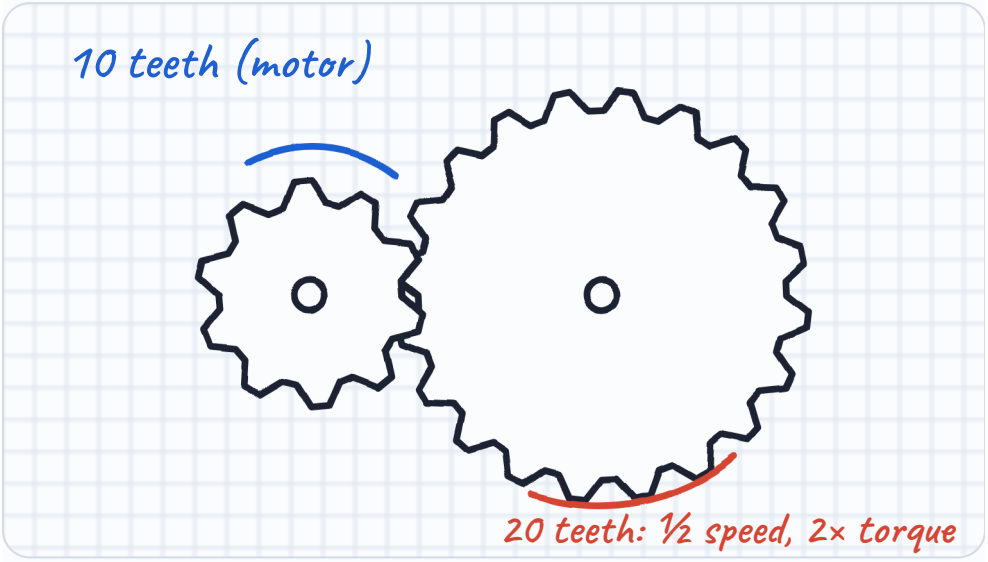


With a moving pulley, the load hangs from two strands of rope, so each strand carries half the weight. You pull with half the force, but you have to pull twice as much rope.

The work you do, force × distance, stays the same. Machines never give energy for free; they let you choose how to spend it.

CONCEPT

Gears: speed or torque



When a small gear drives a big one, the big gear turns slower but with more torque. The ratio is the number of teeth: 10 teeth driving 20 gives half the speed and twice the torque.

A bike's low gear, a drill's gearbox and a car's first gear all use this trade.

QUIZ

1. A 50 kg person stands on a scale. Their weight is about…

- 50 N
- ✓ **490 N**
- 5 N
- 5000 N

W = m × g = 50 × 9.81 ≈ 490 N.

2. 4 kg sits 1.5 m from the pivot. How far from the pivot must 2 kg sit to balance?

- 0.75 m
- 1.5 m
- ✓ **3 m**
- 6 m

4 × 1.5 = 6, so 2 × d = 6 → d = 3 m.

3. A 12-tooth gear drives a 36-tooth gear. The output turns…

- 3× faster
- at the same speed
- ✓ **3× slower with 3× torque**

The ratio is 36 ÷ 12 = 3: one third of the speed, three times the torque.

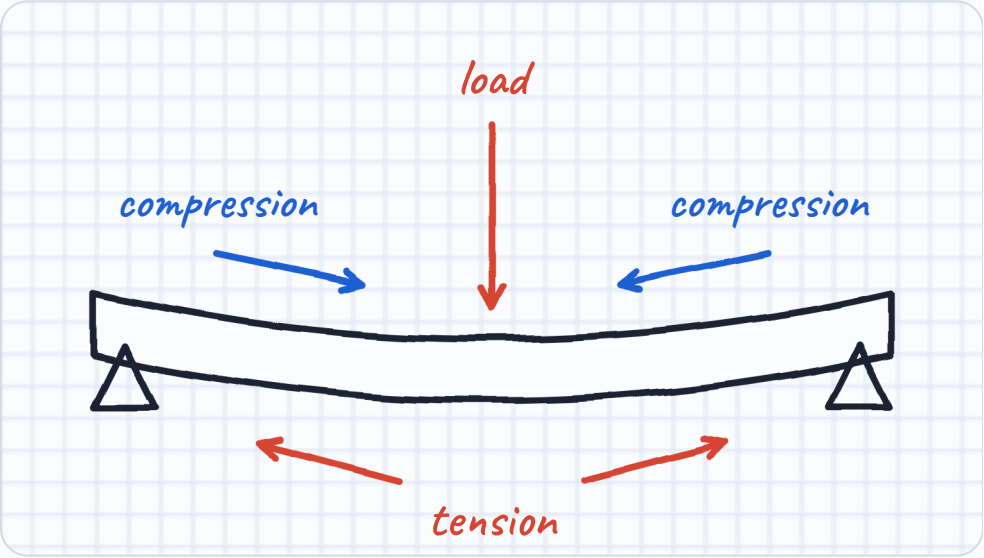
LESSON 2

How structures stand up

Tension, compression and the shapes engineers use to manage them.

CONCEPT

Tension and compression

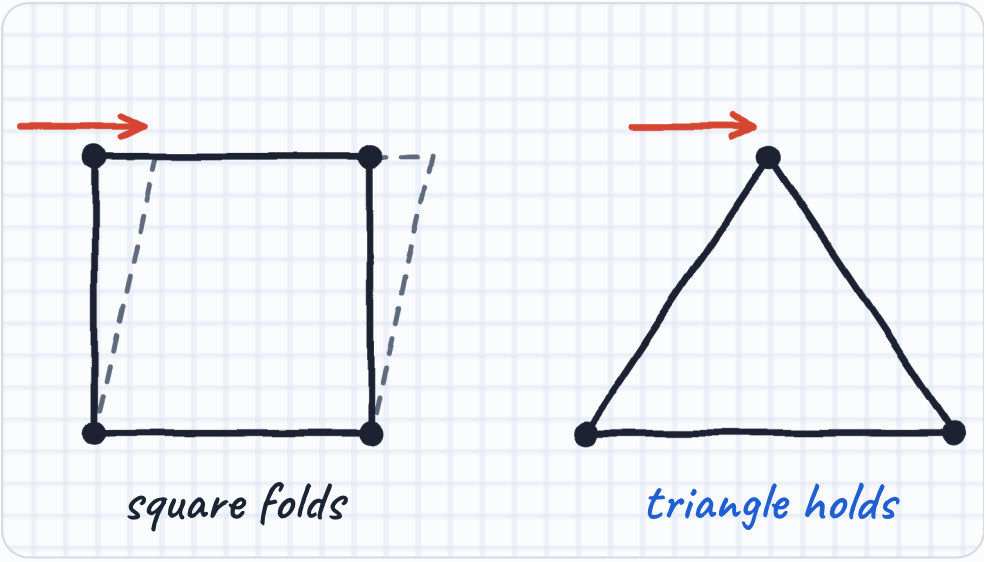


Load a beam in the middle and it bends. The top gets squeezed (compression) and the bottom gets stretched (tension). Somewhere in between is a neutral line that feels neither.

Concrete is strong in compression but weak in tension, which is why beams have steel bars near the bottom.

CONCEPT

Why engineers love triangles

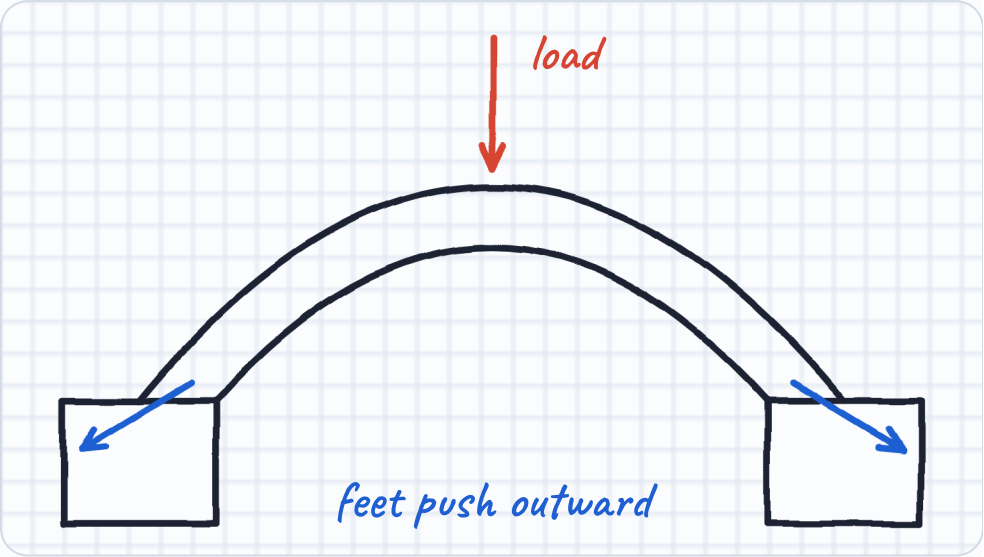


Push the corner of a square frame and it folds into a parallelogram; the joints just rotate. A triangle can't change shape without changing the length of a side.

That's why cranes, roof trusses, pylons and bike frames are made of triangles.

CONCEPT

Arches push outward

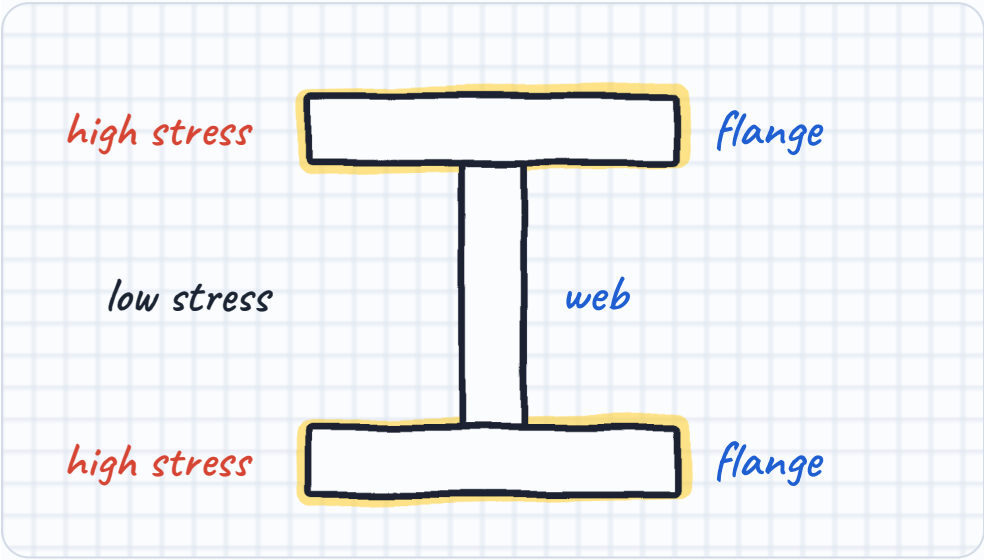


An arch turns a downward load into compression that runs around its curve to the ground. Stone, which is strong in compression, is perfect for it.

The catch is that the feet push outward. Without solid abutments or a tie between the feet, the arch spreads and collapses.

CONCEPT

The I-beam

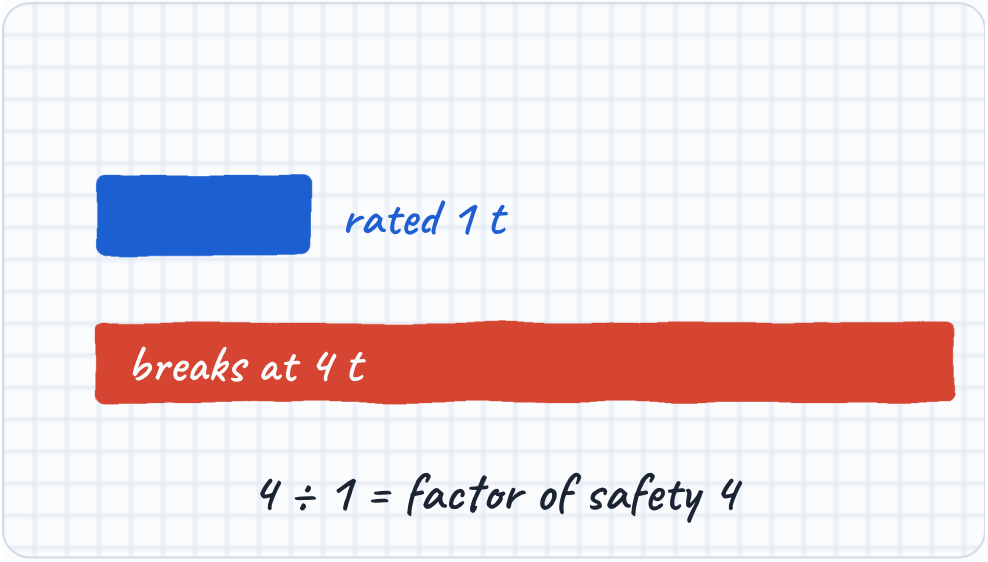


When a beam bends, the most stress is at the top and bottom faces and very little in the middle. An I-beam puts most of its steel in the top and bottom flanges and joins them with a thin web.

You get almost the stiffness of a solid block with a fraction of the weight.

FORMULA

Factor of safety



FoS = failure load ÷ working load

Engineers never design right up to the breaking point. Materials vary, loads get underestimated and things wear out. The factor of safety is how much stronger something is than it needs to be.

Lifting gear often uses 4 or 5; aircraft structures use about 1.5, because extra weight costs fuel and every part is tested closely.

WORKED EXAMPLE

Hook rated for 1 t, breaks at 4 t

4 ÷ 1

factor of safety = 4

QUIZ

1. A simply supported beam carries a load in the middle. Where is it in tension?

- Top face
- ✓ **Bottom face**
- Exactly along the middle

It sags, so the bottom stretches (tension) and the top squeezes (compression).

2. Why are trusses made of triangles?

- They use less steel than squares
- ✓ **A triangle can't deform without a side changing length**
- They look better

Fixed side lengths lock a triangle's angles, so pinned joints alone keep it rigid.

3. Rope with a breaking load of 600 kg, used with a factor of safety of 5. Max working load?

- 3000 kg
- ✓ **120 kg**
- 595 kg

Working load = failure load ÷ FoS = 600 ÷ 5 = 120 kg.

PHYSICS

Physics

Energy, heat and waves in everyday life.

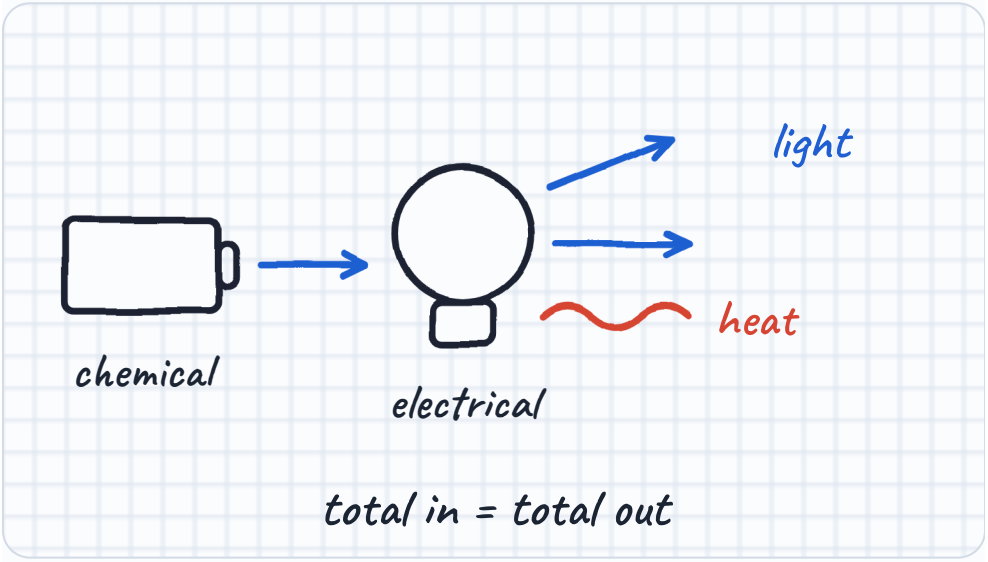
LESSON 1

Energy & heat

Where energy goes, how heat moves, and why heat pumps beat heaters.

CONCEPT

Energy changes form

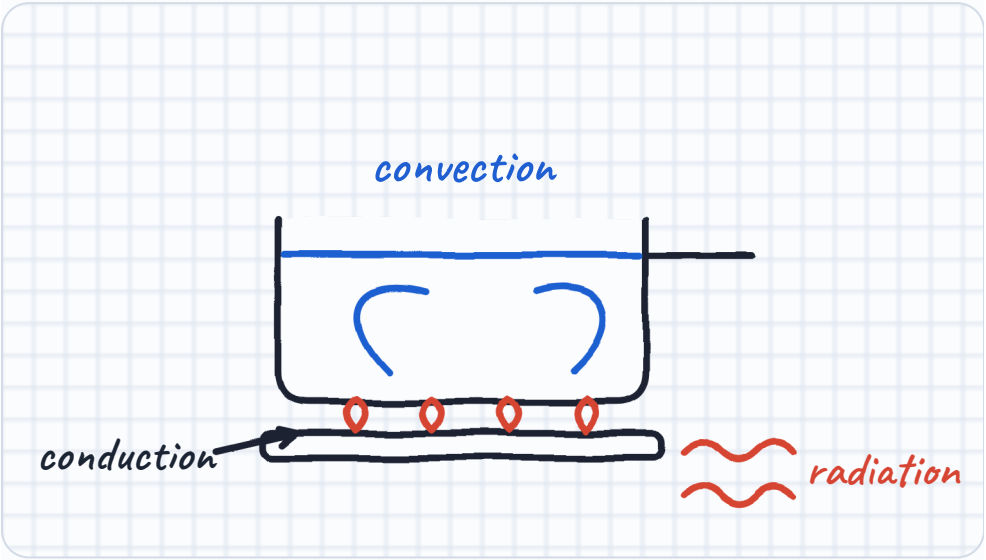


Energy is never created or destroyed, only converted. A battery's chemical energy becomes electrical energy, which a bulb turns into light and heat.

Efficiency is the useful share. An old filament bulb turned about 5% into light; an LED manages several times that, so it runs cooler for the same brightness.

CONCEPT

Three ways heat moves



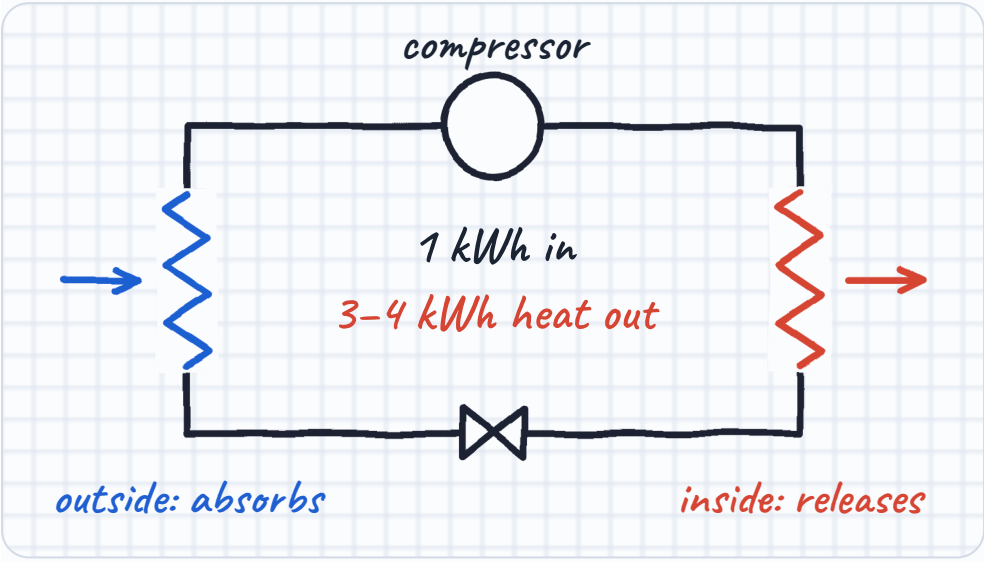
Conduction: heat passes through touching material, from the burner into the pan base.

Convection: warm fluid rises and cool fluid sinks, setting up loops that stir the water.

Radiation: infrared light travels across a gap with no material at all, like the warmth you feel beside a fire.

CONCEPT

Heat pumps move heat

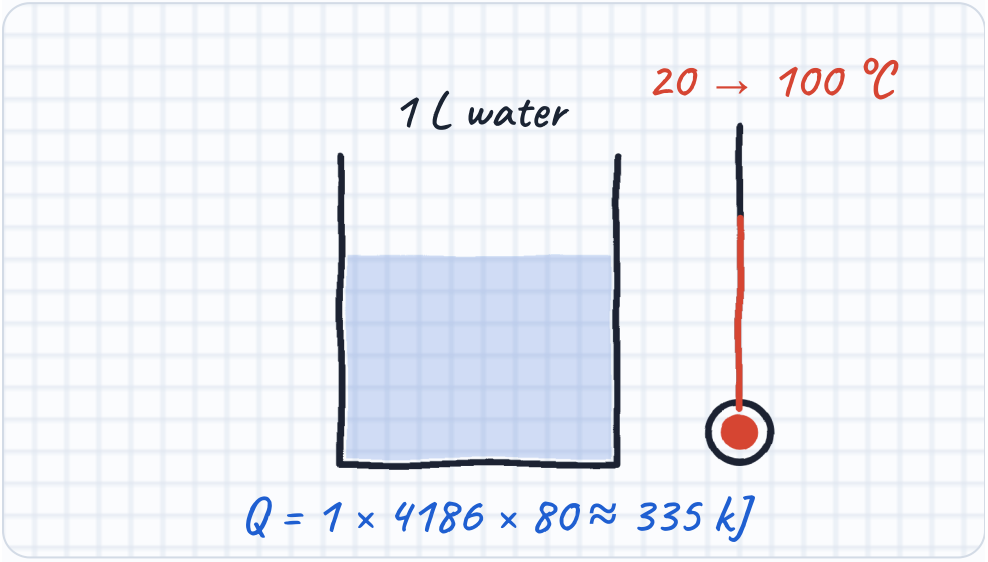


A heat pump doesn't make heat; it moves it. A refrigerant absorbs heat from outside air (even cold air has some), the compressor squeezes it so it gets hotter, and it releases that heat indoors.

Because most of the heat comes from outside, 1 kWh of electricity can deliver 3–4 kWh of heat. That ratio is the COP.

FORMULA

Specific heat



$Q = m \times c \times \Delta T$

The energy to heat something is mass × specific heat × temperature rise. Water's specific heat is high, about 4186 J per kg per °C, which is why it takes a while to boil and why it stores heat so well.

WORKED EXAMPLE

1 L of water from 20 to 100 °C

$1 \times 4186 \times 80 = 335\ kJ$

≈ 0.093 kWh, about 2 cents

QUIZ

1. Which way does heat mainly reach you from the Sun?

- Conduction
- Convection
- ✓ **Radiation**

Space is nearly empty, so only radiation can cross it.

2. A heat pump with a COP of 3 uses 2 kWh of electricity. Heat delivered?

- 0.67 kWh
- 2 kWh
- ✓ **6 kWh**

Heat out = COP × electricity in = 3 × 2 = 6 kWh.

3. Heating 2 kg of water by 10 °C needs about…

- ✓ **84 kJ**
- 8.4 kJ
- 840 kJ

Q = 2 × 4186 × 10 ≈ 83.7 kJ.

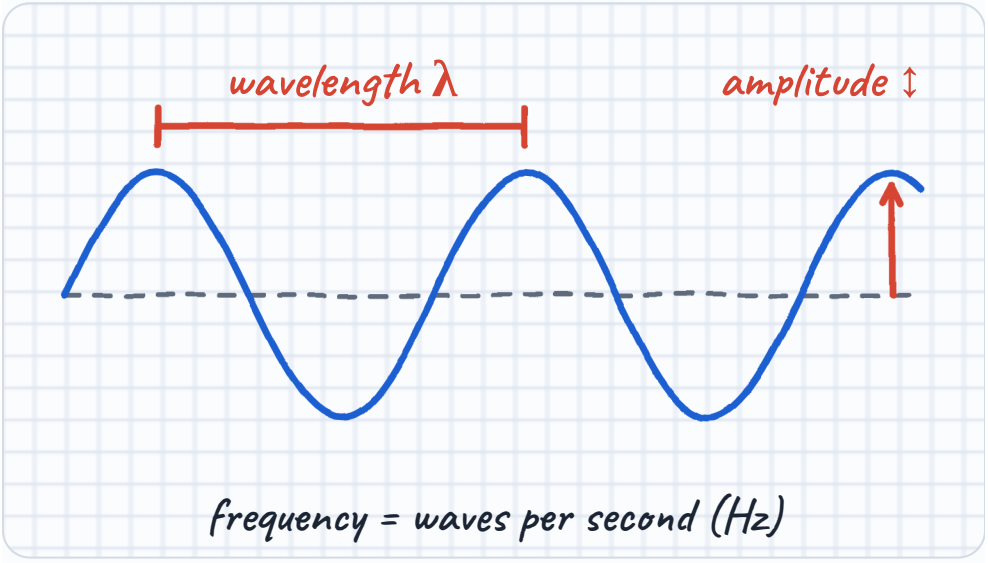
LESSON 2

Waves, sound & light

One equation that covers guitar strings, Wi-Fi and thunderstorms.

CONCEPT

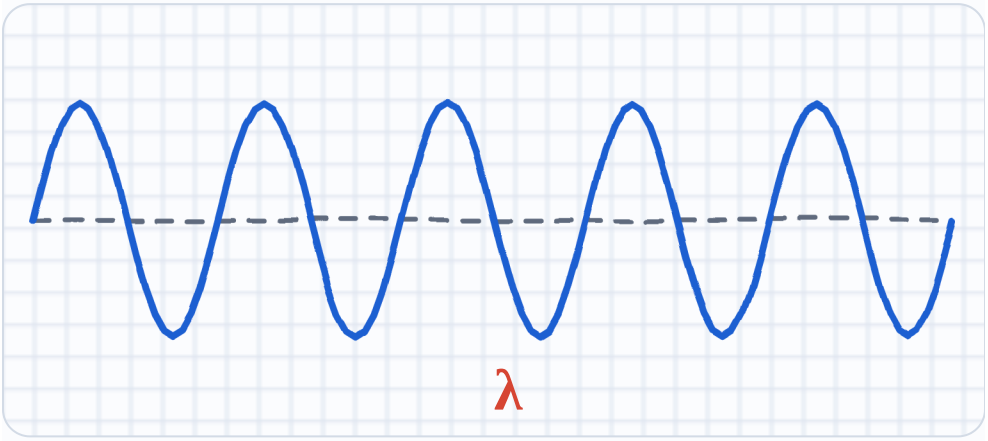
Anatomy of a wave



A wave carries energy without carrying stuff along. Its wavelength (λ) is the distance from one crest to the next. Its amplitude is how tall it is, which sets loudness or brightness. Its frequency is how many crests pass each second, in hertz (Hz).

TRY IT

Try it: how long is a sound?

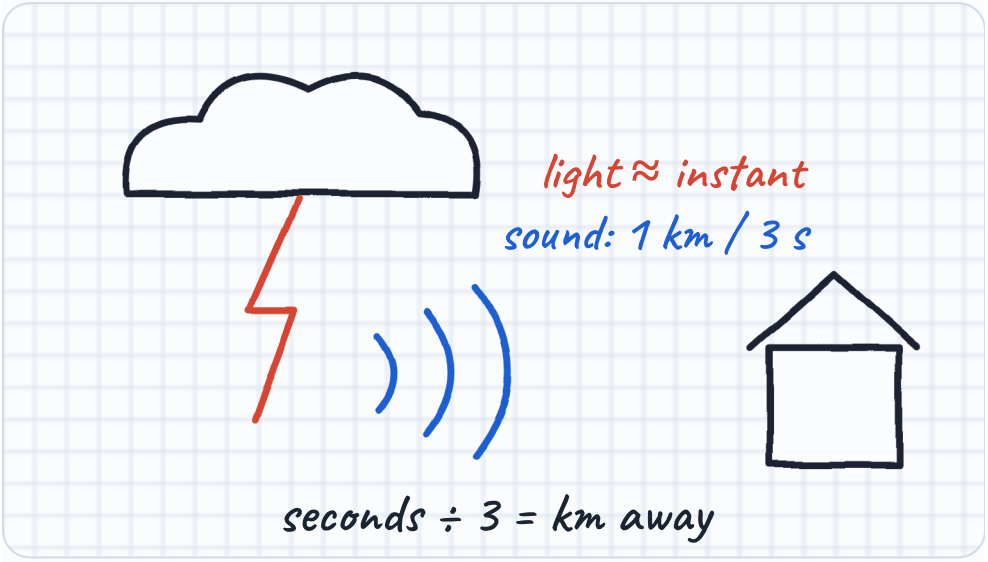


$$v = f \times \lambda$$

Speed equals frequency times wavelength. Sound travels at about 343 m/s in air, so low notes have long waves and high notes have short ones.

FACT

How far away is the storm?

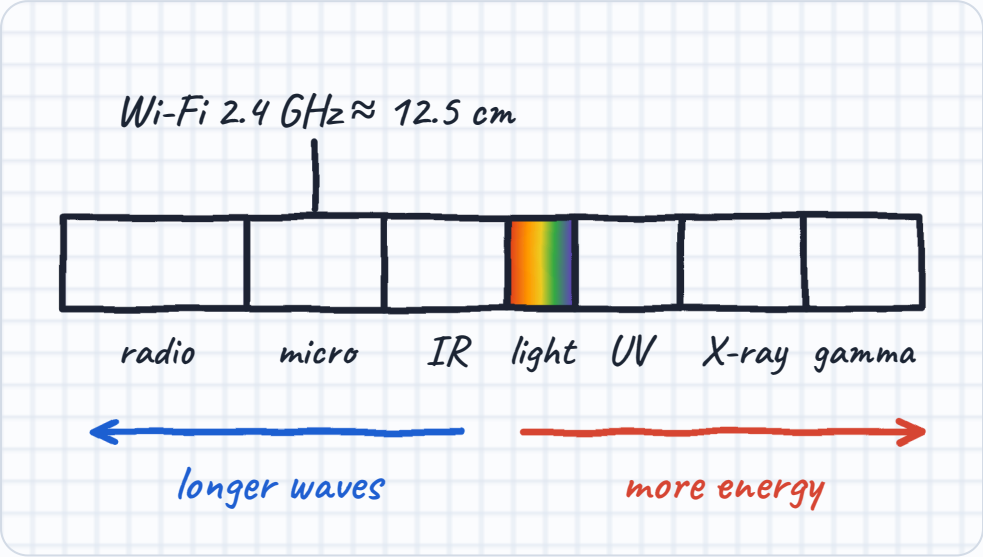


Light from lightning reaches you almost instantly; it travels about 300 000 km every second. Sound crawls along at about 343 m/s, roughly 1 km every 3 seconds.

Count the seconds between the flash and the thunder and divide by 3 to get the distance in kilometres.

CONCEPT

The electromagnetic spectrum



Radio, microwaves, infrared, visible light, ultraviolet and X-rays are all the same thing: electromagnetic waves, all moving at the speed of light. Only the wavelength differs.

Shorter waves carry more energy per photon. That's why UV can burn skin and X-rays pass through it, while radio is harmless. Wi-Fi at 2.4 GHz has a wavelength of about 12.5 cm.

QUIZ

1. A 343 Hz tone in air (343 m/s) has a wavelength of…

- ✓ **1 m**
- 343 m
- 1 cm

λ = v ÷ f = 343 ÷ 343 = 1 m.

2. Thunder arrives 9 seconds after the flash. How far is the strike?

- About 9 km
- ✓ **About 3 km**
- About 1 km

Sound covers about 1 km every 3 s, so 9 s ≈ 3 km.

3. Which carries the most energy per photon?

- Radio
- Visible light
- ✓ **X-rays**

Shorter wavelength means higher frequency and more energy per photon.

HISTORY

History

The turning points of humanity, and Portugal's leap into the open ocean.

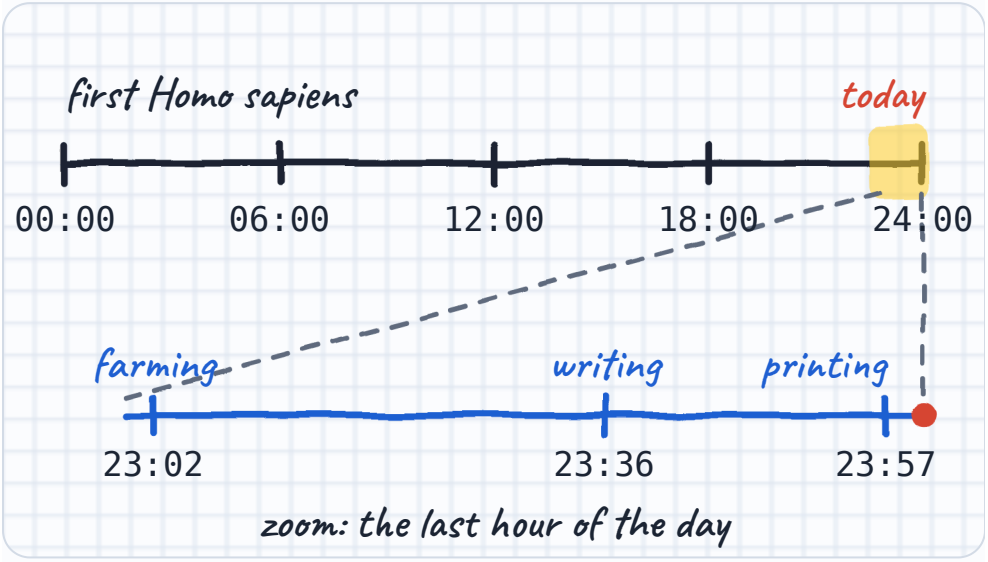
LESSON 1

Turning points of humanity

Five changes that made the modern world, in the order they happened.

CONCEPT

Human history as one day

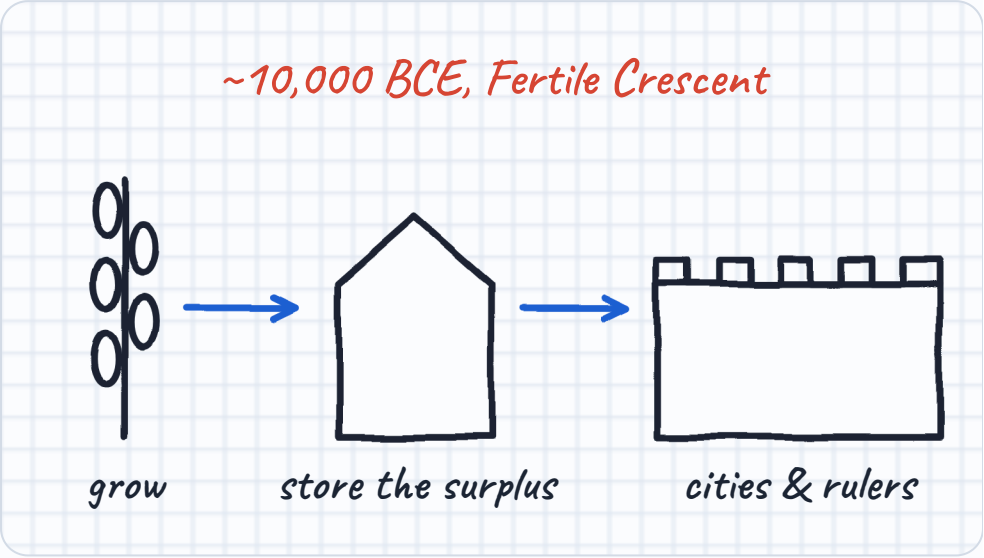


Squeeze the roughly 300,000 years of Homo sapiens into a single day. For almost all of it, everyone lived as hunter-gatherers.

Farming only starts at about 23:02. Writing appears at 23:36. The printing press arrives less than three minutes before midnight, and everything since the steam engine fits in the last minute.

CONCEPT

Farming changed everything

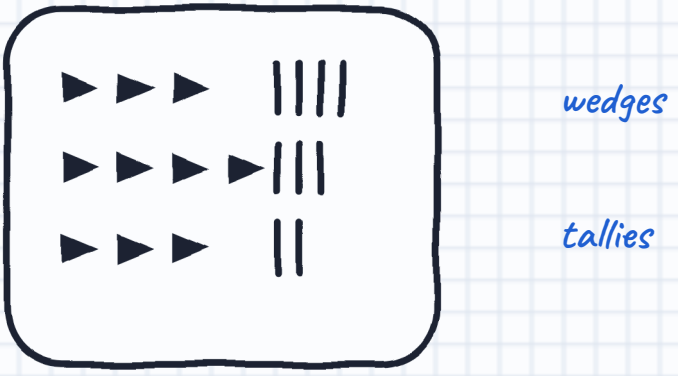


Around 10,000 BCE, people in the Fertile Crescent (today's Iraq, Syria and Turkey) began planting wheat and barley and keeping animals.

Farming made food surpluses possible. Surpluses could be stored, traded and taxed, and they fed people who didn't farm: potters, priests, soldiers and kings. Villages grew into the first cities.

CONCEPT

Writing started as bookkeeping



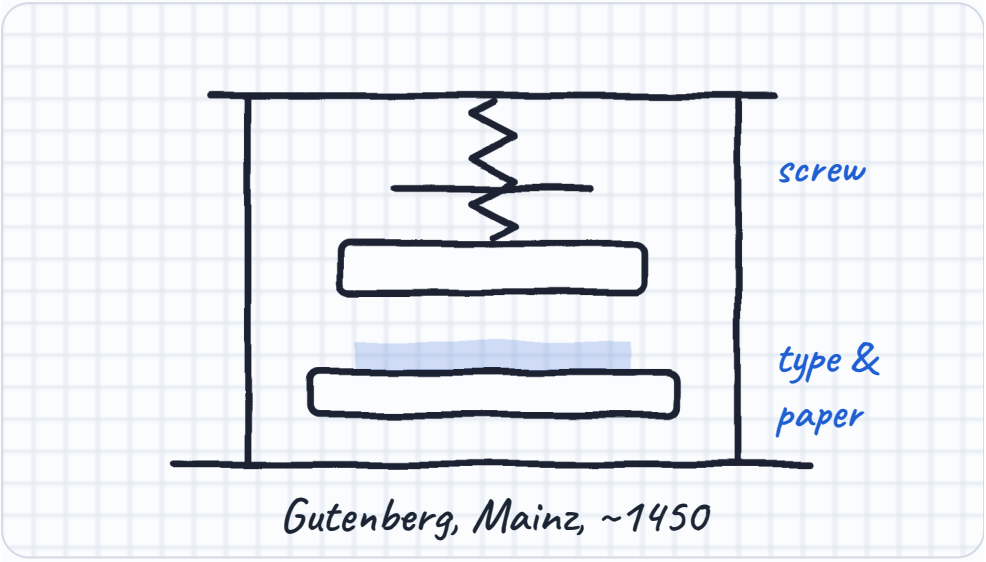
Uruk, ~3200 BCE: barley, sheep, beer

The oldest writing we know comes from Sumer, in today's Iraq, around 3200 BCE. It wasn't poetry. It was accounts: who delivered how much barley, how many sheep, how much beer was paid as wages.

Scribes pressed wedge-shaped marks into wet clay with a reed. We call the script cuneiform, from the Latin for 'wedge'.

CONCEPT

The printing press

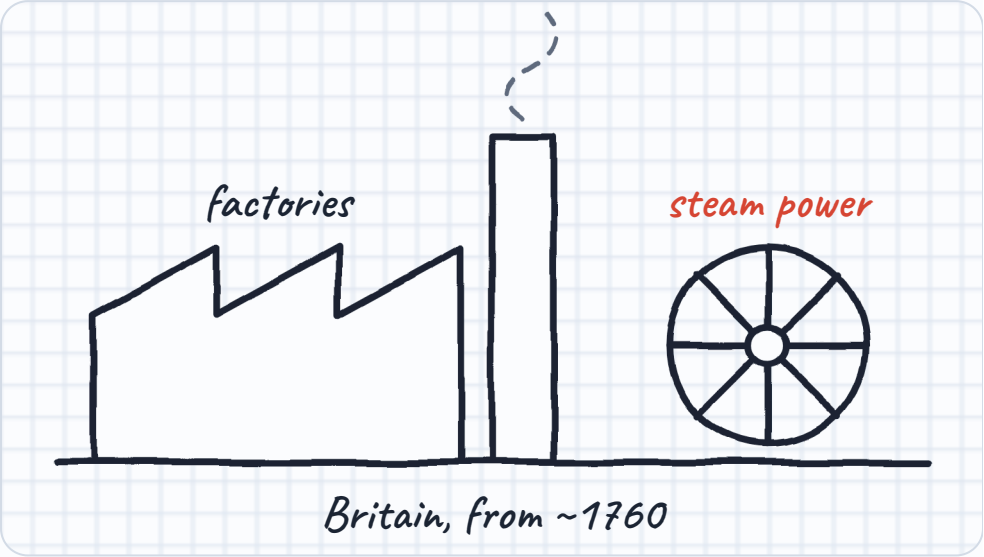


Around 1450 in Mainz, Johannes Gutenberg combined movable metal type, oil-based ink and a press adapted from wine making. A book that took a scribe months could now be printed by the hundred.

Within 50 years millions of books were in circulation. Ideas spread faster than authorities could control them, feeding the Reformation and the scientific revolution.

CONCEPT

The Industrial Revolution



From about 1760 in Britain, machines powered by water and then coal-fired steam engines took over spinning, weaving and mining. Work moved from homes and fields into factories, and people moved into fast-growing cities.

For the first time in history, output per person kept rising decade after decade. It also brought child labour, pollution and the start of large-scale CO_2 emissions.

QUIZ

1. What was most early writing used for?

- Poetry and religious stories
- ✓ **Keeping accounts of goods and payments**
- Letters between kings

The earliest Sumerian tablets are mostly records of grain, livestock and rations.

2. Why did farming lead to cities?

- Farmers liked living close together
- ✓ **Food surpluses could feed people who didn't farm**
- Cities were needed to store tools

Surplus food supported specialists, rulers and traders, which made towns and cities possible.

3. Where and when did Gutenberg's press appear?

- China, ~1050
- ✓ **Mainz, ~1450**
- London, ~1600

Gutenberg worked in Mainz around 1450. (Movable type existed earlier in China and Korea, but his press transformed Europe.)

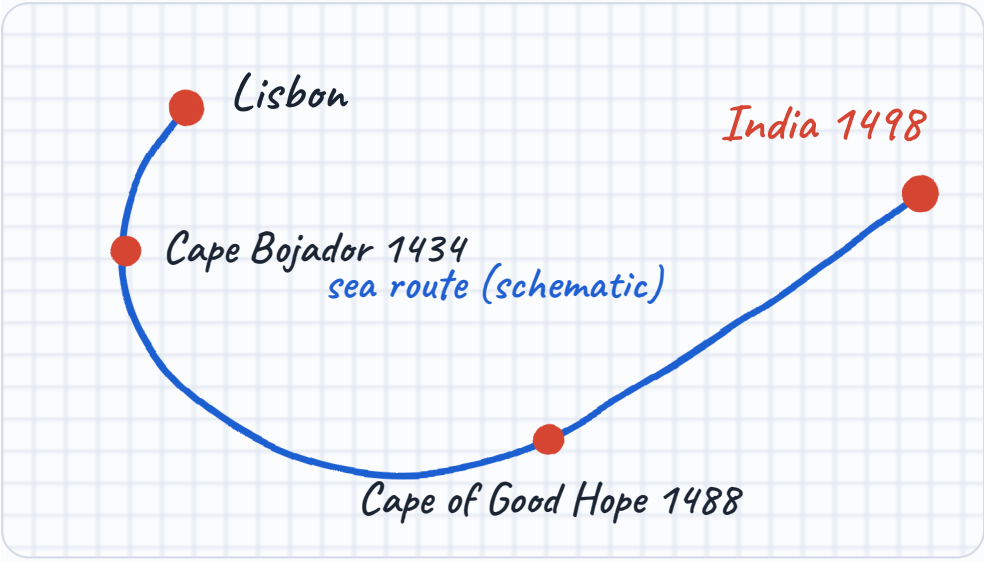
LESSON 2

Portugal and the Age of Discovery

How a small kingdom at the edge of Europe opened the world's sea routes.

CONCEPT

Why Portugal turned to the sea



In the 1400s Portugal was small, poor and boxed in by Castile. The rich trade in spices and silk came overland from Asia and was controlled by Venice and Muslim middlemen.

The Atlantic was the one open door. Prince Henry 'the Navigator' sponsored expeditions down the African coast, step by step, hoping to reach gold, allies and eventually the spices of India by sea.

CONCEPT

The caravel

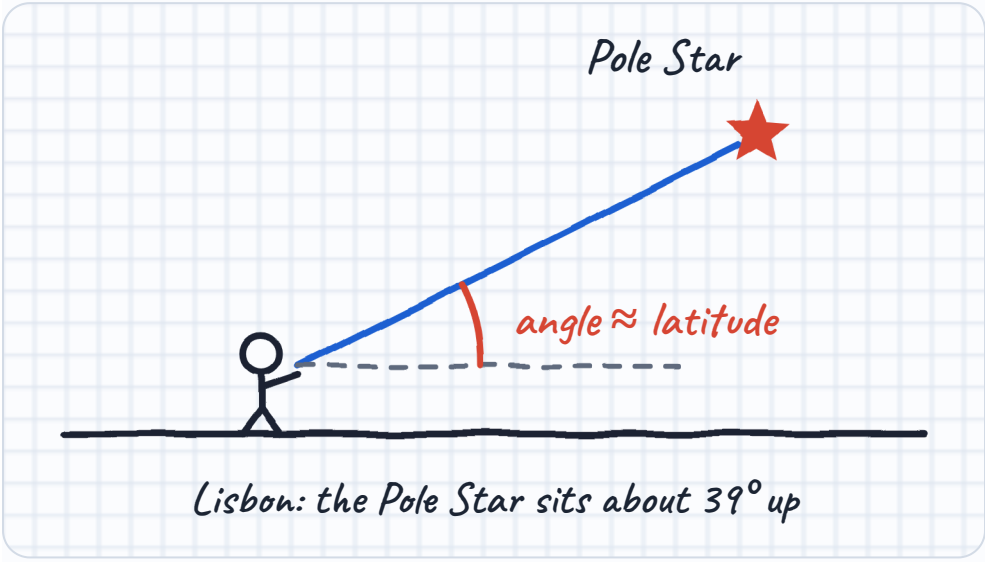


Square sails work beautifully with the wind behind you, but badly against it. The caravel's triangular 'lateen' sails let it sail much closer into the wind, and its shallow hull could explore coasts and rivers.

That mattered: coming home from West Africa meant fighting the prevailing winds. Explorers learned to swing far out into the Atlantic to catch better ones.

CONCEPT

Finding latitude from the stars



On open water there are no landmarks. In the northern hemisphere, the height of the Pole Star above the horizon roughly equals your latitude. Sailors measured it with a quadrant or astrolabe.

Further south the Pole Star disappears, so Portuguese navigators measured the Sun at noon instead and used printed tables to correct for the season.

CONCEPT

The great voyages

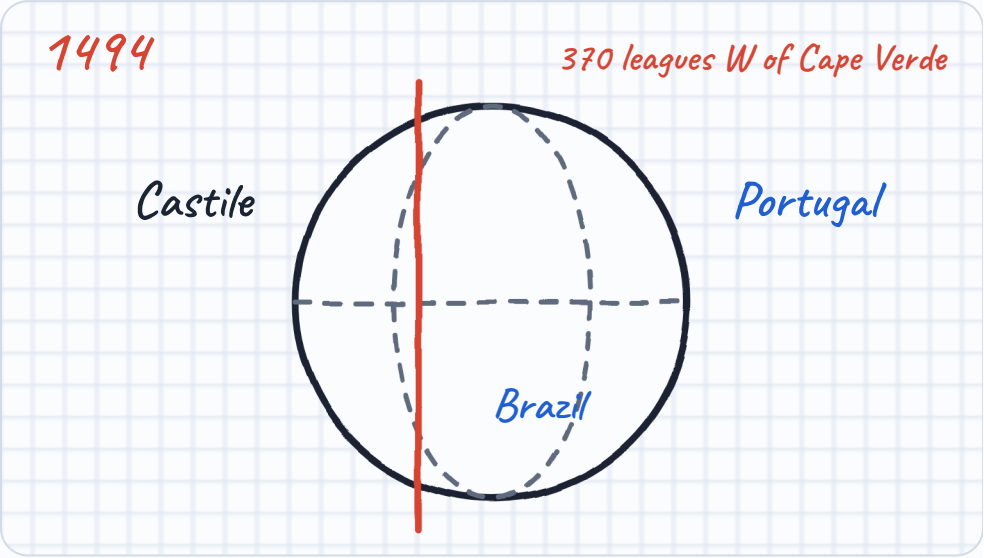


Bartolomeu Dias rounded the southern tip of Africa in 1488. Ten years later Vasco da Gama reached Calicut in India, opening the sea route to the spice trade and making Lisbon one of Europe's richest ports.

In 1500 Pedro Álvares Cabral, heading for India, swung so far west that he reached Brazil. And Fernão de Magalhães (Magellan), a Portuguese sailing for Spain, led the first voyage around the world, completed by Elcano in 1522.

CONCEPT

Splitting the world: Tordesillas



In 1494 Portugal and Castile signed the Treaty of Tordesillas, drawing a line from pole to pole 370 leagues west of the Cape Verde islands. New lands to the west would go to Castile, to the east to Portugal.

The tip of South America that bulged east of the line became Brazil, which is a big part of why around 210 million Brazilians speak Portuguese today.

QUIZ

1. What could a caravel do better than a square-rigged ship?

- Carry more cargo
- ✓ **Sail closer into the wind**
- Sail faster with the wind behind

Its lateen sails let it work into the wind, essential for exploring and getting home.

2. Who first reached India by sea from Europe?

- Bartolomeu Dias
- ✓ **Vasco da Gama**
- Pedro Álvares Cabral

Vasco da Gama reached Calicut in 1498.

3. Why does Brazil speak Portuguese?

- ✓ **Cabral claimed it and it lay on Portugal's side of the Tordesillas line**
- It was bought from Spain
- Magellan settled it

Cabral landed in 1500, and the land fell east of the 1494 treaty line.

WORLD

World

The planet and its people: seasons, time zones, languages and cities.

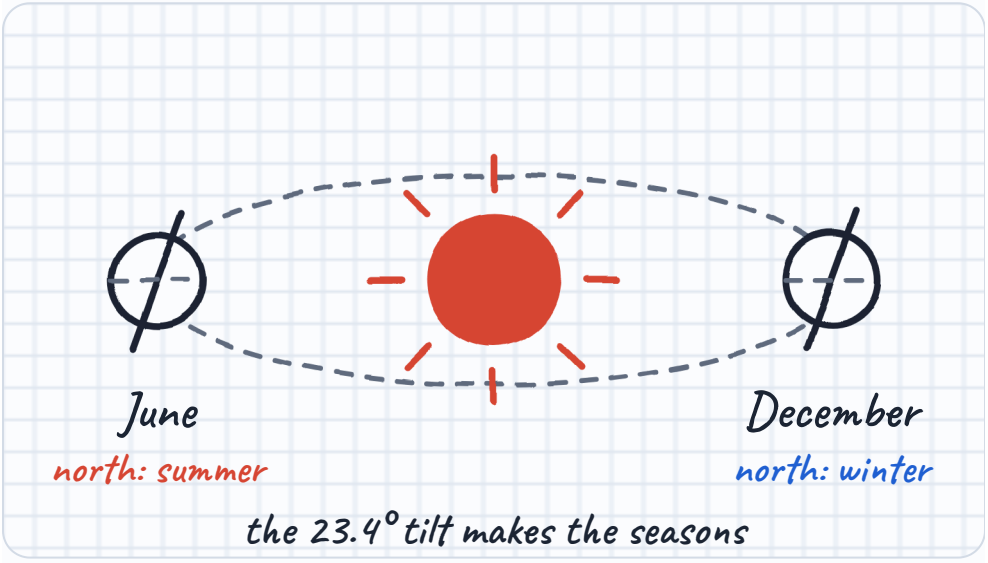
LESSON 1

How the planet works

Seasons, time zones, coordinates and the ocean's central heating.

CONCEPT

Why we have seasons



It isn't about distance: Earth is actually closest to the Sun in early January, in the middle of our winter.

Seasons come from Earth's tilt of about 23.4°. In June the northern hemisphere leans toward the Sun, so sunlight hits it more directly and days are longer. In December it leans away.

TRY IT

Try it: time zones



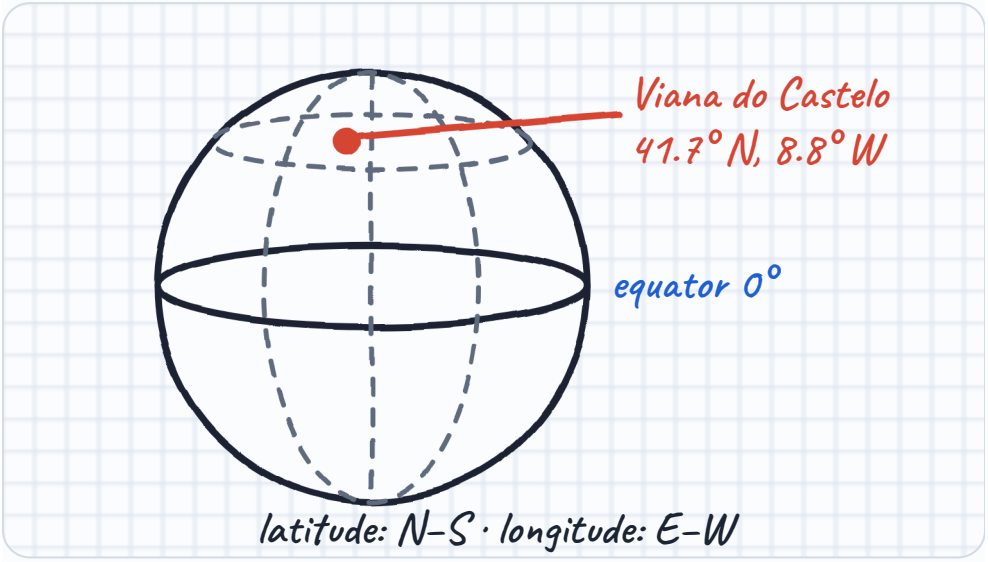
Lisbon

360° ÷ 24 h = 15° per hour

Earth turns 360° a day, so each hour of time difference is about 15° of longitude. Countries round that off to suit their borders, and a few, like India, use half hours. Move Lisbon's clock (winter time) and watch the world follow.

CONCEPT

Latitude and longitude

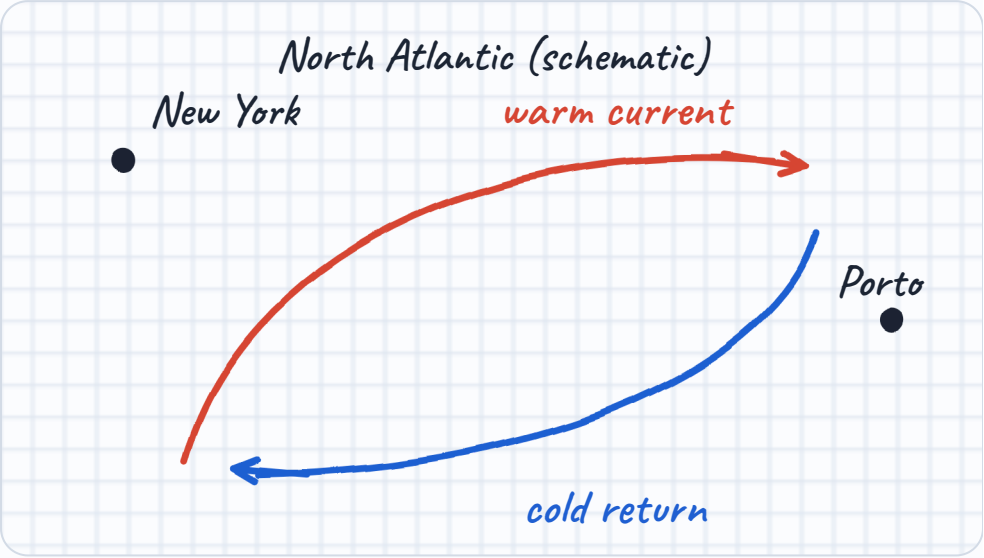


Every place on Earth has two coordinates. Latitude says how far north or south of the equator you are, from 0° to 90°. Longitude says how far east or west of the Greenwich meridian in London, from 0° to 180°.

Viana do Castelo sits at about 41.7° N, 8.8° W.

CONCEPT

The ocean's central heating



Porto and New York sit at almost the same latitude, yet Porto's winters are far milder. Part of the reason is the Gulf Stream and its extension, which carry warm tropical water north-east across the Atlantic.

Westerly winds blowing off that warm ocean bring the heat ashore. Cold water returns south, completing a vast loop.

QUIZ

1. What causes the seasons?

- Earth's distance from the Sun
- ✓ **The tilt of Earth's axis**
- Changes in the Sun's brightness

The 23.4° tilt changes how directly sunlight hits each hemisphere through the year.

2. Two places are 45° of longitude apart. Roughly how many hours apart are their clocks?

- ✓ **3 hours**
- 4.5 hours
- 45 hours

15° per hour, so 45 ÷ 15 = 3 hours.

3. Latitude measures...

- East–west position
- ✓ **North–south position**
- Height above sea level

Latitude runs from 0° at the equator to 90° at the poles.

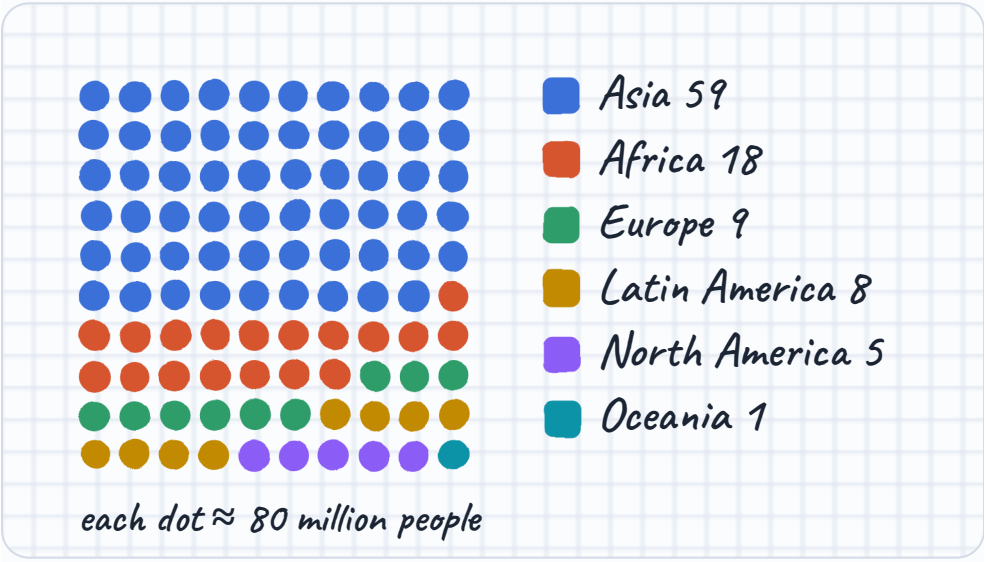
LESSON 2

People and places

Who lives where, what they speak, and why cities are where they are.

CONCEPT

If the world were 100 people

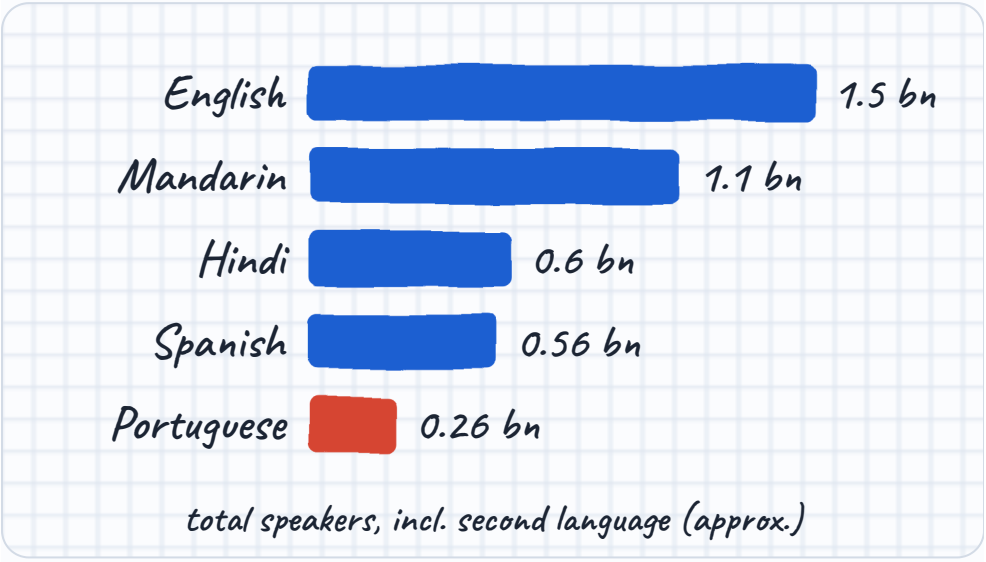


About 8 billion people live on Earth; the UN marked the milestone in November 2022. Shrink that to 100 people and roughly 59 live in Asia, 18 in Africa, 9 in Europe, 8 in Latin America, 5 in North America and 1 in Oceania.

India and China alone hold more than a third of humanity. Africa is the fastest-growing region.

FACT

The most spoken languages

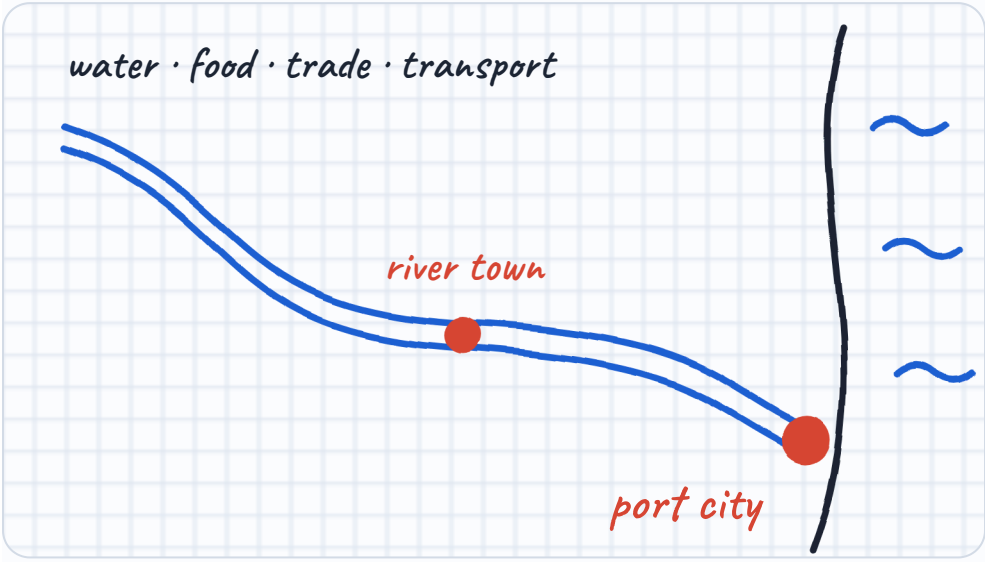


Count everyone who speaks a language, natives and learners together, and English comes first, with Mandarin Chinese second. Count only native speakers and Mandarin leads by a long way.

Portuguese has around 260 million speakers worldwide, most of them in Brazil, which makes it the most spoken language in the southern hemisphere.

CONCEPT

Why cities sit on rivers and coasts



Before railways and roads, water was the motorway. Rivers and sheltered harbours offered drinking water, fertile soil, fish and, above all, cheap transport for trade.

That's why so many great cities stand at a river mouth or a natural harbour: Lisbon on the Tagus, Porto on the Douro, London on the Thames, Shanghai at the mouth of the Yangtze.

FACT

How many countries are there?



The usual answer is 195: the 193 members of the United Nations plus two observer states, the Vatican City and Palestine. Some places, like Taiwan and Kosovo, govern themselves but aren't UN members, so counts differ.

The largest is Russia, at about 17 million km². The smallest, Vatican City, is under half a square kilometre.

QUIZ

1. Which region is home to most of the world's people?

- Africa
- Europe
- ✓ **Asia**

Roughly 59 out of every 100 people live in Asia.

2. Which language has the most native speakers?

- English
- ✓ **Mandarin Chinese**
- Spanish

Mandarin leads on native speakers; English leads when learners are included.

3. Why did so many big cities grow on rivers and coasts?

- Cooler climate
- ✓ **Water, food and cheap transport for trade**
- Easier to defend from the air

Water routes were the cheapest way to move goods and people for most of history.